

Eco-efficiency outcomes of mergers and acquisitions in the European electricity industry

Evgenii Monastyrenko

Paris School of Economics - Paris 1 Panthéon Sorbonne University, 106-112, Boulevard de l'Hôpital, 75013 Paris, France



ARTICLE INFO

JEL:

F21
G34
L25
L94
D24

Keywords:

Electric power industry
Mergers and acquisitions
Eco-efficiency
Data envelopment analysis
Fractional regression model
Carbon dioxide emissions

ABSTRACT

The European electricity industry was recently liberalized. In response, there was a surge of mergers and acquisitions (M & As). This study addresses the effects of M & As on the eco-efficiency of European electricity producers in 2005–2013. The environmental production technology comprises CO₂ emissions as an undesirable output. I compute eco-efficiency using data envelopment analysis (DEA) and the Malmquist-Luenberger productivity index, which are both based on radial directional distance function. I observe a decreasing trend in average eco-efficiency, which contradicts the previously documented moderate efficiency gains of liberalization. The effects of M & As are isolated using second-step fractional regression. The domestic horizontal M & As, which are systematically regulated by the European Commission, have no impact. Although one cross-border horizontal deal has a same-year effect of roughly –3%, this effect becomes approximately +1.5% over a two-year timespan. Vertical domestic M & As have a short-run negative impact of 3.6% that does not persist over time. Vertical international M & As reduce the eco-efficiency by 2.1% two years after their completion. Limited evidence suggests that the conglomerate deals are at least not harmful. The policy implication is that the merger regulation should be based on DEA eco-efficiency measures. Regulators should devote more attention to cross-border M & As and particularly to vertical deals.

1. Introduction

Europe is the third-most important electricity market after Asia and North America. In 2013, the European Union (EU) represented 19.1% of worldwide energy-generation capacity and produced 16.4% of global electricity. In the early 1990s, the European energy market was highly regulated, as each state aimed to ensure its own energy security. In most of these states, the installed generation capacity largely overlapped with actual needs. Because utilities were forced to use internal resources instead of low-cost alternatives, they did not diversify their fuel sources, had low operational and corporate flexibility and relied on long-term forecasting (Serrallés, 2006). Thus, energy producers had no incentives to become efficient. The liberalization of the European electricity sector started in 1996 with Directive 96/92/EC whose objective was the establishment of a common, competitive EU market. The three declared aims were consumer choice in electricity suppliers, third-party access to the transmission and distribution system, and unbundling distribution networks from generation.

As a consequence of this liberalization, EU electricity producers have faced the challenges of drastically increased competition and turbulent operational environments. These market players initiated an intense wave of mergers and acquisitions (M & As), which allowed them

to swiftly readjust their allocations of distribution and production capacities. Prior to liberalization, energy producers functioned entirely within national borders and, hence, were commonly referred to as “national champions”. These firms pursued cross-border M & As to increase their cost efficiency, incorporate foreign managerial expertise and gain access to distribution networks. Thus, liberalization has induced M & As, a high share of which are cross-border deals. In 2010, the global energy industry completed many deals, second only to the financial sector (Schmid et al., 2012). European electricity producers have greatly contributed to that wave. In 2011, they accounted for approximately 35% of global energy intakes.

This study addresses the period governed by the second and third EU electricity directives. The second directive (2003/54/EC) has produced the most liberalization-related policy changes. The most significant changes were the establishment of independent national regulators and free entry into electricity generation. The directive further strengthened the vertical unbundling of distribution and transmission activities from generation and retail. The third directive (2009/72/EC) introduced ownership unbundling of energy production from transmission. The European Commission further promoted the independence of national regulators and established the Agency for the Cooperation of Energy Regulators (ACER).

E-mail address: evgenii.monastyrenko@univ-paris1.fr.

<http://dx.doi.org/10.1016/j.enpol.2017.04.030>

Received 9 November 2016; Received in revised form 3 March 2017; Accepted 16 April 2017

Available online 04 May 2017

0301-4215/ © 2017 Elsevier Ltd. All rights reserved.

This paper belongs to a narrow branch of the M & A literature that aims to assess post-merger alterations in firm-level efficiency. Several studies on electricity generation and distribution are worth mentioning. The only previous attempt to address international M & As in the energy sector was conducted by Berry (2000), who did not find any impact. Becker-Blease et al. (2008) revealed the financial losses caused by mergers between US electric utilities. They also found that integration with gas activities and diversification into new geographic areas are detrimental to both market and operating performance. Kwoka and Pollitt (2010) applied data envelopment analysis (DEA) and assessed the performance outcomes of M & As in the US electricity market between 1994 and 2003. They did not find any evidence that M & As improve cost performance. Bagdadioglu et al. (2007) and Çelen (2013b) predicted efficiency outcomes of electricity market reforms in Turkey. Çelen (2013a) exploited a two-stage DEA approach and revealed the determinants of the efficiency of Turkish electricity distribution companies.

To the best of my knowledge, only three published studies have assessed firm-level post-merger outcomes in the European electricity industry, and two of these studies are devoted to the initial stage of the liberalization. Bednarczyk et al. (2010) addressed 37 cross-border M & As between bidders from Western industrialized countries and targets from Central and Eastern Europe over the 1995–2005 period. Their event study revealed a small, positive impact on target shareholder wealth in the short run. Datta et al. (2013) studied 156 mergers completed from 1990 to 2006 among electricity, gas, water and telecommunication firms. These authors revealed that combined EU utilities experienced statistically significant losses over both the short and long run. Saastamoinen et al. (2017) studied merger gains in the electricity distribution sector of Norway. These authors argue that merger outcomes significantly vary with respect to the regulatory model.

This paper investigates the outcomes of 129 M & As that were completed by the 15 largest European electricity producers between 2004 and 2013. All of these firms are multinationals that combine fossil, nuclear and renewable sources for electricity generation. I employ DEA to compute firm-level eco-efficiency and isolate the causal impacts of M & As. DEA is a nonparametric mathematical technique that is frequently applied in performance evaluation and operational research in the financial sector. I contribute to a growing strand of energy and environmental (E & E) studies exploiting DEA. Recent applications in the energy sector include Yang and Pollitt (2009), Zhang et al. (2013), Arabi et al. (2014), Atkinson and Tsionas (2016) and Duan et al. (2016).

I compute eco-efficiency scores using an output-oriented DEA model based on a radial directional distance function (DDF). I employ installed capacity and total operational expenditure as inputs and generated electricity and carbon dioxide (CO₂) emissions as good and bad outputs, respectively. I address the panel nature of the dataset with two alternative approaches. The window DEA approach assumes the absence of technological progress. I assess the obtained DEA eco-efficiency scores using a fractional quasi-maximum likelihood estimator with correlated random effects. As an alternative, I compute the Malmquist-Luenberger productivity index (MLPI), which has recently become a common measure in intertemporal efficiency evaluation. In this case, I use OLS with time and firm fixed effects as the second-stage estimator.

The results suggest that in the short run, domestic mergers reduce eco-efficiency by an average of 3.35%. The effects of cross-border deals appear to be weaker (−2.06%). In total, 93 cross-border M & As are analyzed, representing 72.1% of the sample. This study is the second since Bednarczyk et al. (2010) to differentiate between horizontal and non-horizontal energy mergers. I find that carefully regulated domestic horizontal mergers do not have a statistically significant impact. Cross-border horizontal mergers hamper eco-efficiency in the short run but stimulate it two years after completion. An interesting and somewhat

new result is that vertical mergers are detrimental to eco-efficiency. More precisely, domestic vertical mergers have a short-run negative impact of 3.63%, but this impact does not persist over time. Cross-border vertical mergers dampen eco-efficiency by 2.1% over a two-year horizon. Regulatory authorities should adapt efficiency-promoting merger regulation using this study as a reference.

The paper proceeds as follows. In Section 2.1, the pattern of M & As among EU electricity producers is characterized. I describe the DEA model in Section 2.2, and I explain the comprehensive second-stage fractional regression analysis in Section 2.3. Then, I report the computed eco-efficiency scores in Section 3.1. The effects of cross-border and domestic mergers and acquisitions are quantified in Section 3.2. I further distinguish among the eco-efficiency outcomes of horizontal, vertical, and conglomerate M & As in Section 3.3. Finally, Section 4 summarizes the research and discusses the policy implications.

2. Data and methodology

2.1. M & As

In this study, I use panel data for the 15 largest European electricity producers over the 2005–2013 period. In 2005, the firms of interest represented 54.2% of installed EU capacity and produced 55.1% of total electricity. In 2013, they generated 62.3% of all European energy and represented 53.2% of the region's capacity. The complete list of firms is presented in Table A.1.

The data on M & As were extracted from the Thomson Reuters SDC Platinum database. The sample is composed of 129 M & As. Hereinafter, I employ the term “mergers”.¹ Note that all deals in the sample are cases where one large EU electricity producer acquires a smaller entity. Mergers included in the sample satisfy the following conditions: First, the date of completion was between January 2004 and December 2013; second, the acquisition was realized by one of the studied energy firms rather than by an affiliate, associate or subsidiary company; and third, the acquirer's stake was initially below 50%, but after completion of the acquisition, the stake was 50% or higher.

The total sample is composed of 93 cross-border (72.1%) and 36 domestic (27.9%) deals. I classify a merger as a cross-border deal if the headquarters of the target and acquirer are not located in the same country. In some cases, the firms of interest finalized both international and domestic mergers in the same year. The number of deals of each type is reported separately for each firm-year observation. A timeline of M & A activity is shown in Fig. 1. European energy firms merged most intensively in 2006 and 2008, whereas the highest share of international mergers was observed in 2005. In 2004 and 2010, nearly as many domestic as international mergers occurred.

I further differentiate among horizontal, vertical and conglomerate mergers (see Fig. 2). Whereas horizontal mergers occur between electricity utilities, the vast majority of targets in vertical mergers are separate power plants that co-generate heat and electricity or use alternative energy sources. Other vertical targets operate in the extraction of fossil fuel or in the production of turbines and generators. The targets in conglomerate mergers are not related to any stage of energy generation or to energy distribution. I report the number of deals in each category in Table 1. Disaggregation at the sector level is presented in the Appendix (Table A.2).

¹ For instance, the European Merger Regulation (European Commission, 2004, 2008) is dedicated exclusively to mergers. In the M & A literature, the terms “merger” and “acquisition” are often used synonymously. However, they have different definitions. A merger occurs when two entities of similar size are combined into one new entity. An acquisition occurs when a larger firm purchases a smaller firm or entity. In practice, mergers per se are quite rare.

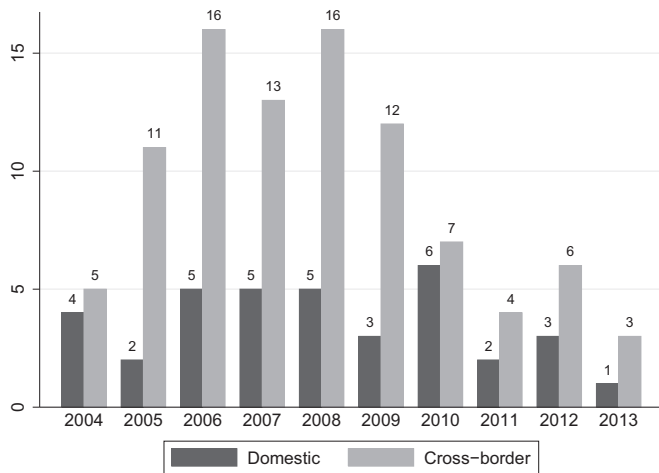


Fig. 1. Evolution of domestic and cross-border M&As.

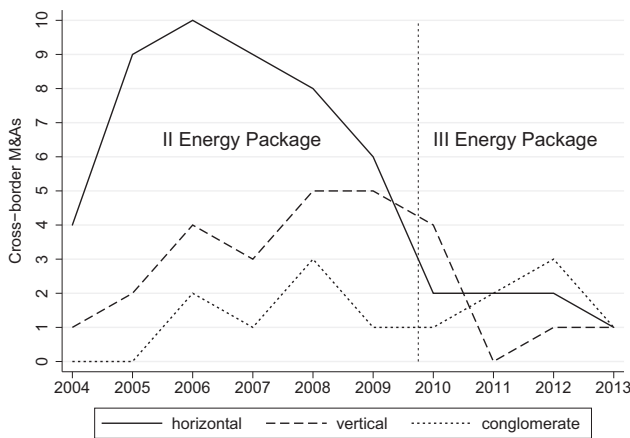


Fig. 2. Repartition of M&As by types.

Table 1
Types of mergers.

Types	Domestic	Cross-border	Targets' principal activities	U.S. SIC codes
Horizontal	17	53	Energy utility	4911, 4931, 4939
Vertical	9	26	Electricity generators, gas and oil extraction	1221, 1311, 1351, 3621, 499A
Conglomerate	10	14	Non-related to electricity generation and distribution	1522, 1731, 2813, 3433, 3674, 4922, 4923, 4924, 4941, 5172, 6512, 6719, 7389, 8711, 8742, 8748

Notes: Table reports classification of M&As in the sample.

2.2. DEA

DEA has been frequently applied to assess the performance of various industries. This nonparametric technique was introduced by Charnes et al. (1978). It is based on the observed inputs and outputs of decision-making units (DMUs). An important advantage over parametric techniques is that DEA does not require specification of a cost function; nor are strong assumptions about the exact form of the production function needed.² DEA has recently gained a widespread attention in E&E studies.³

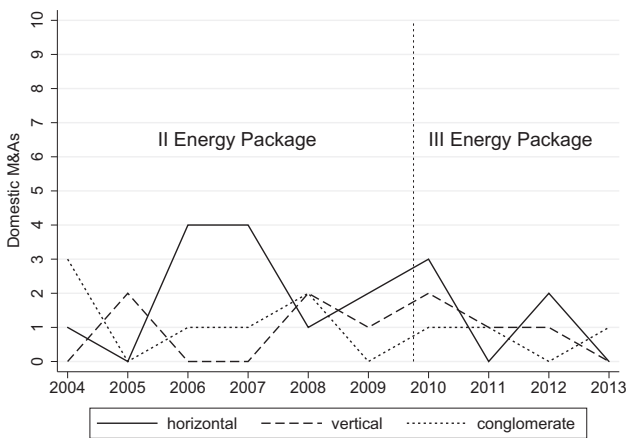
Assume that the market consists of $j=1, \dots, N$ electricity firms. They make use of the input vector $x \in \mathbb{R}_+^m$ to produce desirable output

² See, e.g., Jamasb and Pollitt (2003) for a comparison of DEA to corrected ordinary least squares (COLS) and stochastic frontier analysis (SFA) approaches. They estimated the efficiency of regional European electricity distribution utilities.

³ See the comprehensive literature surveys in Zhou et al. (2008), Zhang and Choi (2014) and Sueyoshi et al. (2017).

$y^d \in \mathbb{R}_+^s$ and undesirable output $y^u \in \mathbb{R}_+^j$. The results of DEA efficiency studies are sensitive to the choice of inputs and outputs. I rely on previous E&E studies, data availability and the specificity of electricity generation to make these choices.

In this study, the vector of inputs x is composed of two inputs: installed production capacity (in MW) and total operational expenditures (in millions of euros). Jamasb and Pollitt (2003) argued that operational expenditures and installed capacity are common inputs in studies on electricity distribution. Vaninsky (2006) and Sueyoshi et al. (2010) incorporated operating expenses into an assessment of the US electric power generation industry. Pérez-Reyes and Tovar (2009) summarized the DEA components of 14 studies of electricity distribution. The majority of them employ installed capacity, together with operational costs, to proxy for physical capital. The survey of Arabi et al. (2014) revealed the same pattern in plant-level eco-efficiency studies of energy producers. Saastamoinen et al. (2017) employed total costs as the single input in an efficiency assessment of the Norwegian electricity distribution industry.



Labor is another potential input that is exploited in a significant share of DEA efficiency studies. I do not consider this input due to data limitations. The studied European electricity producers are large multinationals that operate in several areas alongside electricity generation.⁴ Explicit reports of the number of employees engaged in power production occurs only in a few cases. The inclusion of total labor as an input would lead to the underestimation of eco-efficiency for firms that have more non-generation activities. It is worth noting that labor is a particularly appropriate input for efficiency studies of electricity generation at the plant level (e.g., Yang and Pollitt, 2009; Färe et al., 2014).

The desirable output y^d is generated electricity (TWh), the principal physical output in the energy sector. The quasi-totality of previous studies on the eco-efficiency of generators targets the amount of generated power (e.g., Arabi et al., 2014). I consider total CO2

⁴ Examples include heat generation, gas production and power distribution.

emissions (tons) produced during electricity generation as the undesirable output y^u . The most common undesirable outputs considered in previous DEA studies of electricity generation are CO₂, sulfur dioxide (SO₂) and mono-nitrogen oxides (NO_x). Sözen et al. (2010) and Sueyoshi et al. (2010) introduced all of these as separate undesirable outputs. I include neither SO₂ nor NO_x emissions because these data are unavailable. Moreover, similar recent studies of energy generation (e.g., Bai-Chen et al., 2012; Zhang et al., 2013; Duan et al., 2016) employ CO₂ emissions as the only bad output. In 2013, the public electricity and heat-generation EU sector was responsible for approximately 28.9% of total EU28 greenhouse gas emissions (Guerreiro et al., 2014).

I source the inputs and outputs from the energy utility data provided by Enerdata.⁵ For some observations, I retrieve data on CO₂ emissions and electricity production from PricewaterhouseCoopers (2013). The definitions and sources of the DEA components are summarized in Table 2.

The multi-output production function can be described by:

$$T = \{(x, y^d, y^u): x \text{ can produce } (y^d, y^u)\} \quad (1)$$

Färe et al. (1989) formulated assumptions of weak disposability and null-jointness that are imposed on the production technology T . These assumptions can be represented as follows:

$$\text{If } (x, y^d, y^u) \in T \text{ and } 0 \leq \theta \leq 1, \text{ then } (x, \theta y^d, \theta y^u) \in T \quad (2a)$$

$$\text{If } (x, y^d, y^u) \in T \text{ and } y^u = 0, \text{ then } y^d = 0 \quad (2b)$$

Weak disposability (Eq. (2a)) requires that reduction of the undesirable output y^u is costly in terms of the proportional reduction of desirable output y^d . The null-jointness assumption (Eq. (2b)) states that undesirable outputs are not avoidable. The only way to produce zero CO₂ emissions is to stop electricity generation.

The production function in Eq. (1) is defined conceptually, and it must be specified for further empirical estimation. This study employs a nonparametric DEA piecewise linear framework, as in Zhou et al. (2012) and Zhang et al. (2013). The environmental production technology with constant returns to scale (CRS) can be represented in generalized form as follows:

$$T = \{(x, y^d, y^u): \sum_{n=1}^N \lambda_n x_{mn} \leq x_m, m = 1, \dots, M, \\ \sum_{n=1}^N \lambda_n y_{sn}^d \geq y_s^d, s = 1, \dots, S, \\ \sum_{n=1}^N \lambda_n y_{jn}^u = y_j^u, j = 1, \dots, J, \\ \lambda_n \geq 0, n = 1, \dots, N. \} \quad (3)$$

The DDF for eco-efficiency estimation was introduced by Chung et al. (1997). The principal advantage of this approach is that it allows simultaneous increases in desirable outputs and decreases in undesirable outputs and/or inputs. Färe and Grosskopf (2004) argue that traditional radial efficiency models represent a special case of the DDF, which can be articulated as follows:

$$\vec{D}(x, y^d, y^u, \vec{g}) = \sup\{\beta: (y^d, y^u) + \beta \vec{g} \in T(x)\} \quad (4)$$

The directional vector $\vec{g}$ defines the orientation in which the outputs are scaled.⁶ I use the vector $\vec{g} = (y^d, -y^u)$ which is frequently

Table 2
DEA input and output components.

Variable	Definition	Data sources
Capacity	Total installed capacities involved in electricity generation (MW)	Enerdata and reports
TOPEX	Total operational expenditures (M€)	Thomson One
PowerGen	Physical amounts of generated electricity (TWh)	Enerdata and reports
Emissions	Total CO ₂ emissions from electricity generation (tons)	Enerdata, reports and PricewaterhouseCoopers (2013)

Notes: Table describes inputs and outputs used in the setup of DEA model.

applied in E&E studies.⁷ The feasible increase in the amount of produced electricity is intended to be consistent with a simultaneous reduction of CO₂ emissions. I assume that the inputs remain constant. Thus, my model is output-oriented. The assumption that the electricity firms could not alter their installed capacities is realistic since almost all of the relevant electric plants were operating throughout the whole period of interest.

The “classical” radial DDF has an underlying assumption that desirable and undesirable outputs can be adjusted at the same rate β . A potential issue with radial DDF is that it might overestimate efficiency when the slacks are non-zero. Non-radial slack-based DEA models are good alternatives for radial ones.⁸ The major difference relative to the radial approach is that inputs, desirable outputs and undesirable outputs can be adjusted at different rates that are predefined by the directional vector.

Sahoo et al. (2011) summarizes the discussion of the choice between radial and non-radial functions. The superior mathematical properties of the radial function and the simplicity of applications in management should be noted. Moreover, the radial approach has been more frequently employed (see Zhang and Choi, 2014). Recent applications include, e.g., Beltrán-Estevé and Picazo-Tadeo (2015) and Beltrán-Estevé and Picazo-Tadeo (2017). However, Sueyoshi and Yuan (2016) note that determining which approach is preferable in terms of efficiency measurement is not straightforward.

In this study, the environmental production technology is estimated using A radial DDF model. I reshape the broadly defined undesirable-measure DEA framework (Eq. (3)) so that it includes the radial directional eco-efficiency measure β_{CRS}^* . I assume an increase in the desirable output and a reduction in the undesirable output given constant quantities of inputs. Following Aparicio et al. (2013) and Färe et al. (2014), I introduce an upper bound on the optimal level of CO₂ emissions, which is equal to the maximum observed value of this undesirable output. Such an approach prohibits counter-intuitive cases in which additional electricity could be produced with a reduction of emissions.

The final optimization program is as follows:

⁵ Enerdata is an independent research and consulting agency that specializes in energy market forecasting, policy assessment and efficiency evaluation. Their discontinued Energy Market Intelligence dataset is the source of data until 2010, which includes financial data and production highlights for the major European energy producers. Information since 2010 is extracted from Power Plant Tracker. This database records the characteristics of power plants worldwide. I employ the “Company Dashboard” tool, which contains aggregate information on the assets and footprints of the main power companies.

⁶ Atkinson and Tsionas (2016) proposed to use Bayesian methods to estimate optimal firm-specific directions for each particular input and output. They demonstrated an application based on the dataset of 77 US electricity utilities. Note that the data requirements include prices of inputs.

⁷ See Zhang and Choi (2014) for a comprehensive overview of the evolution of DDF through 2013. These authors discovered only two input-oriented E&E studies.

⁸ One of the first applications of such a model is Zhou and Ang (2008) in which the energy efficiency of 21 Organization for Economic Co-operation and Development (OECD) countries was measured. In addition, Zhang et al. (2013) studied the cross-sectional efficiency of Korean power generators using a meta-frontier non-radial DDF.

$$\begin{aligned}
\vec{D}(x, y^d, y^u) &= \max \beta \\
s. t. \quad &\sum_{n=1}^N \lambda_n x_{nm} \leq x_m, m = 1, \dots, M, \\
&\sum_{n=1}^N \lambda_n y_{sn}^d \geq y_s^d + \beta y_s^d, s = 1, \dots, S, \\
&\sum_{n=1}^N \lambda_n y_{jn}^u \geq y_j^u - \beta y_j^u, j = 1, \dots, J, \\
&\max\{y_j^u\} \geq y_j^u - \beta y_j^u \\
&\lambda_n \geq 0, n = 1, \dots, N, \beta \geq 0
\end{aligned} \tag{5}$$

This study employs a panel dataset of 15 energy producers over 9 years. Most DEA studies utilize cross-sectional datasets, but they fail to address changes in efficiency over time. The so-called window analysis approach can be used with panel data. The DMUs are regarded as separate in each period.⁹ I selected the entire 9-year period as the window. The underlying assumption is that the technological frontier does not move within the window. The farther an electricity producer is from the best-practice frontier, the less efficient it is. Eco-efficiency is achieved when $\beta_{CRS}^* = 0$, whereas $\beta_{CRS}^* > 0$ signals inefficiency. For the sake of simplicity, I compute the direct eco-efficiency scores as follows: $efficiency_{it} = 1 - \beta_{it}$. The values are on an interval (0,1]. A firm has a score of unity in a given year if, at that moment, it is the most efficient of the firms over the entire period. Values below 1 indicate inefficiency.

To examine the robustness of the obtained results, it is worth relaxing the aforementioned assumption of the absence of technological progress. The Malmquist productivity index (MPI) is a common non-parametric approach that addresses intertemporal productivity change. It can be decomposed into technical change (frontier shift) and efficiency change (position relative to the frontier). These components are also referred to as the technical and allocative efficiencies. Recent E & E studies employ the Malmquist-Luenberger productivity index (MLPI) that is based on the DDF and incorporates both desirable and undesirable outputs. The MLPI was introduced by Chung et al. (1997) and has been further developed, e.g., in Zhou et al. (2010) and Zhou et al. (2012).

In this study, I make use of the refined version of the MLPI proposed by Aparicio et al. (2015). The authors argue that when decomposing the overall efficiency, the allocative efficiency should be derived first. In order to maintain comparability with the window DEA approach, I use the same set of inputs, outputs and constraints.¹⁰ For the sake of interpretability, I follow the fixed-reference Malmquist model (Berg et al., 1992), which captures the change in the efficiency of a DMU relative to a previous or baseline period. The year 2005 is chosen as the baseline period. I compute the MLPI for each firm-year observation over the 2006–2013 period, and the values reflect changes in eco-efficiency in the current period relative to the 2005 values. MLPI values above unity indicate a rise in eco-efficiency, whereas values below unity indicate a decline. I further decompose the aggregate MLPI into technical efficiency change (MLTEC) and technical change (MLTC).

2.3. Regression analysis

I conduct a two-stage DEA, which allows the computation of eco-efficiency values and the isolation of the causal influences of contextual factors. Yang and Pollitt (2009) summarizes a number of the advantages of two-stage DEA. First, no strong prior judgment about the direction of the relationship between efficiency and the regressors is required. Second, both categorical and continuous variables can be included without increasing the sample size. Finally, the analysis is relatively straightforward to apply and the results easy to interpret compared to three- or four-stage approaches. Pombo and Taborda

(2006), Pérez-Reyes and Tovar (2009) and Çelen (2013a) apply two-stage DEAs to study the efficiency of electricity distribution companies.

The appropriate selection of the second-stage estimator should build on the properties of the obtained measures of eco-efficiency. In the principal DEA approach, I compute the eco-efficiency scores that belong to the unity interval. The linear estimation models are not pertinent for explaining variation in the proportional or fractional response variables, since the predicted values can exceed unity (see, e.g., Yang and Pollitt, 2009; Ramalho et al., 2011). Note that the logit and probit models are not the best alternatives (see, e.g., Murteira and Ramalho, 2016). McDonald (2009) has demonstrated that the Tobit estimator is not appropriate either. The two classes of fractional regression models are practically exploited. Papke and Wooldridge (1996) introduce a quasi-maximum likelihood estimator (QMLE) based on the Bernoulli quasi-likelihood function. It is employed to estimate conditional mean models. An alternative is a fully parametric model that is conventionally estimated by maximum likelihood (ML).¹¹

I use a correlated random effects probit QMLE that has been previously applied to fractional outcome models, e.g., in Papke and Wooldridge (2008) for balanced panels and Wooldridge (2009) for unbalanced panels.¹² The values of the Malmquist-Luenberger index can exceed unity. In the corresponding second-stage models, I employ an OLS estimator with time and firm fixed effects.

The following general form describes the estimation strategy:

$$\begin{aligned}
DEAScores_{it} = & \alpha_0 + \alpha_1 NumM\&As_{it} + \beta_1 GenToSales_{it} + \beta_2 TwoMergers_{it} + \\
& + \beta_3 TwoCBMergers_{it} + \gamma_1 K/S_{it} + \gamma_2 I/K_{it} + \gamma_3 Y/S_{it} + \\
& + \gamma_4 Leverage_{it} + \gamma_5 Leverage_{it}^2 + \gamma_6 CurrentRatio_{it} \\
& + \gamma_7 CurrentRatio_{it}^2 + \\
& + \sum \delta_m Location_{it} + \sum \delta_t^{2013} \delta_t Year_t + \varepsilon_{it}
\end{aligned} \tag{6}$$

The variables of interest are the numbers of mergers of a certain type completed in each year.¹³ First, I estimate the effects of all types of mergers. Next, I distinguish between domestic and cross-border M & As, and I further differentiate them with respect to acquirer-target industry relatedness as horizontal, vertical and conglomerate mergers. Each estimated mode l contains time averages for all explanatory variables because their inclusion controls for correlations between the regressors and time-invariant unobservables. The definitions of the control variables are summarized in Table 3. The financial and managerial controls are computed based on data from the Orbis by Bureau van Dijk and Thomson One by Thomson Reuters databases.¹⁴

Recently merged firms must integrate their management systems and practices with the systems and practices of their targets. In the context of the electricity industry, the production structure might be adapted to a new fuel mix. Furthermore, experienced acquirers should be better able to efficiently negotiate the terms of a deal. Firms that

¹¹ See the recent contribution of Murteira and Ramalho (2016) for a comprehensive survey of both conditional mean and parametric models. They also propose several specification tests.

¹² Fixed effects estimators have recently attracted widespread attention. However, they often fail to produce unbiased estimates in non-linear models because of the incidental-parameter issue. Greene (2004) demonstrated that different varieties of ML fixed effects models are inconsistent in cases with finite samples. The existing approaches to correcting this bias (e.g., Hahn and Newey, 2004; Fernández-Val, 2009) often require long panels. Furthermore, strong assumptions of stationarity and serial independence are needed, which drastically limits the practical applications of these adjustments.

¹³ A possible alternative approach estimates separate categories of M & As. It consists of introducing categorical dummies indicating the type of acquisition. I refrain from using this approach because of the complexity inherent in interpreting its results. Indeed, ambiguity can arise from “mixed” categories in which, e.g., both cross-border and domestic mergers are completed.

¹⁴ Three firms report their financial highlights in currencies other than the euro: CEZ A.S. uses Czech crowns, E.ON SE uses US dollars, and Vattenfall AB uses Swedish crowns. I obtain the values in euros using the year-averaged exchange rates provided by the European Central Bank.

⁹ Examples of relevant applications include Wang et al. (2013), Kao and Liu (2014) and Rahman et al. (2016).

¹⁰ The optimization programs for both window DEA and MLPI are solved using the Data Envelopment Analysis Toolbox for MATLAB (Alvarez et al., 2016).

Table 3
Control variables.

Variable	Definition	Data sources
GenToSales	The share of own generation in total amounts of sold energy	Enerdata and reports
TwoMergers	Dummy that takes unity value if a firm completed M & As both in the period of interest and one year ago	SDC Platinum
TwoCBMergers	Dummy that takes unity value if a firm is involved in cross-border M & As both in the period of interest and one year ago	SDC Platinum
Location	Installed capacities (MW) in separate countries: Germany, France, UK, Italy, Portugal, Spain, Czech Republic, Poland, Benelux and Nordic EU countries	Enerdata and reports
K/S	The ratio of long term tangible assets (property, plant, and equipment) to net sales	Thomson One
Y/S	The ratio of operating income to net sales	Thomson One
I/K	The ratio of capital expenditures to tangible long term assets (property, plant, and equipment)	Thomson One
Leverage	Financial leverage computed as the ratio of total liabilities to shareholders' equity	Thomson One
Current ratio	The ratio of current assets to current liabilities	BvD Orbis

Notes: Table represents definitions and sources of included control variables.

have recently experienced an M & A may be able to undertake these adjustments more quickly and at lower costs.¹⁵ I control for merger experience via the TwoMergers dummy, which is equal to one if a firm has finalized M & As in two consecutive years.

Cross-border merger experience should impact eco-efficiency differently. Firms must integrate entities that function under different macroeconomic conditions and in another regulatory environment. Markides and Ittner (1994) stated that acquirers with prior international experience should gain more from cross-border mergers. Such firms are better able to use their experience to integrate and efficiently manage the acquisition process. However, an energy utility that has previously engaged human and financial resources in cross-border readjustments could be more vulnerable during a merger in the next year. I introduce the TwoCBMergers dummy, which is equal to one if a utility underwent an international merger in the current and the prior year.

Mainstream theories of industrial organization and economics predict that vertical integration should promote firm efficiency. Kwoka (2002) revealed that large cost savings are associated with coordination between energy generation and distribution. Kwoka and Pollitt (2010) confirmed this result and demonstrated that vertical integration positively influences the overall performance of electricity producers. In this study, I control for vertical integration using the GenToSales variable, which is equal to the share of the distribution output generated by the utility itself.

A firm's efficiency is largely influenced by its management. An enormous amount of the literature (see, e.g., Jose et al., 1996) is related to the principal-agent problem.¹⁶ Exposure to the principal-agent problem is proportional to the degree of information asymmetry between the involved parties. Physical assets can be relatively easily observed and monitored. This vulnerability to the principal-agent problem is proxied by the capital-to sales (K/S) ratio. This ratio is equal to long-term tangible assets (property, plants, and equipment) divided by net sales. The more a firm uses hard capital, the easier it is for shareholders to monitor the firm.

Physical capital is a key production factor in the electricity industry. Electricity producers make intensive capital investments in two cases, with the main case being the construction and launch of new capacity. Alternatively, production capacity may be replaced or renovated, e.g., to comply with ecological norms. Such alterations are expected to strongly influence the eco-efficiency of electricity producers. In the present study, I control for the ratio of capital expenditures to tangible long-term assets (I/K), as in Himmelberg et al. (1999). These authors argue that firms with higher investment rates have more opportunities

to implement discretionary projects, and consequently, these firms grow more quickly.

The inelastic nature and seasonality of energy demand are important particularities of the electricity industry and provide energy firms with great opportunities to exploit their market power. Producers can use two mechanisms to control energy prices (Helman, 2006). The first mechanism is physical withholding, which consists of holding electricity generation below productive capacity. The alternative is to employ financial withholding by setting an excessively high price. Therefore, market power is expected to influence firm-level eco-efficiency, which depends on the production capacity in use and on the generated electricity. I follow Himmelberg et al. (1999) and control for the ratio of operating income to net sales (Y/S).

Leverage is an important measure of financial sustainability and reflects the extent to which a firm's activities are financed by creditors. The trade-off theory of the optimal capital structure describes the choice of leverage as the equilibrium between the costs and benefits of debt (see, e.g., Kraus and Litzenberger, 1973). The performance of low-leverage firms is a positive function of the debt-to-capital ratio because agency problems have been overcome. The performance of high-leverage firms deteriorates as the amount of leverage increases for two reasons. First, they have fewer incentives to further invest in capital. Second, servicing such debts requires excessive cash flows (Coricelli et al., 2012). In the present study, I control for leverage using linear and quadratic terms, where leverage is defined as the ratio of total liabilities to shareholder equity.

A firm's daily activities require a sufficient level of working capital, and the amount of cash on hand must be sufficient to cover current debts and other obligations. This indicator is often referred to as liquidity. An inverted U-shaped relationship between liquidity and profitability has been confirmed empirically by Baños-Caballero et al. (2012) and Aktas et al. (2015).¹⁷ In this study, the current ratio is computed as current assets divided by current liabilities. I introduce linear and quadratic terms of this ratio as a second-stage, right-hand-side variable.

The functioning of energy firms is subject to national regulations. Despite substantial efforts by the European Commission, EU members still differ in their level of control over electricity production. A good example is nuclear phase-out in Germany. Tax legislation is another dimension of cross-country variation. Therefore, some of the variation in the eco-efficiency of firms can be attributed to their locations. I proxy for the locations of energy firms using the installed generation capacities of the following counties: Czech Republic, France, Italy, Poland, Portugal, Spain and the United Kingdom. I sum (by firm) the

¹⁵ Kamien and Zang (1993) found that the synergetic gains are amplified in serial acquisitions. These authors demonstrated theoretically that such deals allow a firm to monopolize the sector. Bertrand and Betschinger (2012) argued that serial acquirers could face larger organizational challenges and higher financial constraints.

¹⁶ Top management team members (agents) could make decisions that would enhance their wealth but largely diverge from the objectives of the principals (shareholders), leading to poor firm performance.

¹⁷ Firms with more liquidity are better protected against the risk of default. However, a trade-off exists between liquidity and operating performance. Whereas profitability increases with lower investments in working capital (Jose et al., 1996; Wang, 2002; García-Teruel and Martínez-Solano, 2007), if the level of working capital is too low, the firm will lose its ability to adjust sales in accordance with the actual market situation (see, e.g., Wang, 2002).

Table 4
Correlations and descriptive statistics.

	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12
1 Capacity	39204.3	34670.5	1											
2 TOPEX	31864.8	29933.6	0.52*	1										
3 PowerGen	159.2	153.8	0.97*	0.48*	1									
4 Emissions	56621068.8	49796889.1	-0.09	0.17*	-0.11	1								
5 TwoMergers	0.24	0.43	0.14	0.06	0.12	0.08	1							
6 TwoCBMergers	0.17	0.38	0.17*	0.14	0.15	0.09	0.81*	1						
7 GenToSales	0.77	0.20	0.37*	-0.09	0.38*	-0.03	-0.04	-0.03	1					
8 K/S	1.29	0.60	0	-0.54*	0.02	-0.51*	0	-0.03	0.46*	1				
9 I/K	0.12	0.067	-0.15	0.35*	-0.12	0.35*	0.04	0.06	0.03	-0.34*	1			
10 Y/S	0.15	0.087	-0.33*	-0.56*	-0.3*	-0.21*	-0.05	-0.08	0.34*	0.71*	-0.09	1		
11 Leverage	2.90	1.95	0.52*	0.15	0.57*	0.07	0	-0.03	0.04	-0.12	-0.05	-0.35*	1	
12 CurrentRatio	1.10	0.24	0.02	0.10	0.07	-0.05	0	0.06	0.04	-0.03	0.04	0.01	-0.03	1

Notes: Table reports correlation matrix and descriptive statistics. * indicates significance at 5% level.

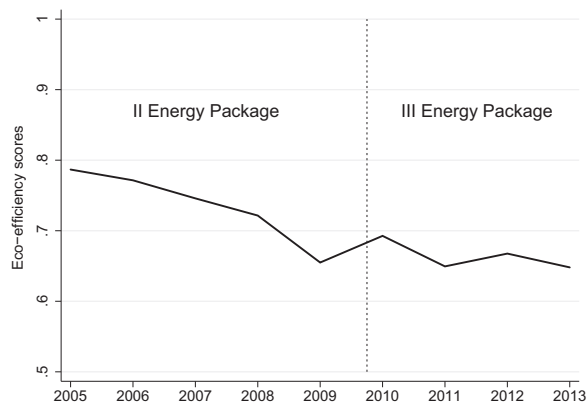


Fig. 3. Average eco-efficiency DEA scores.

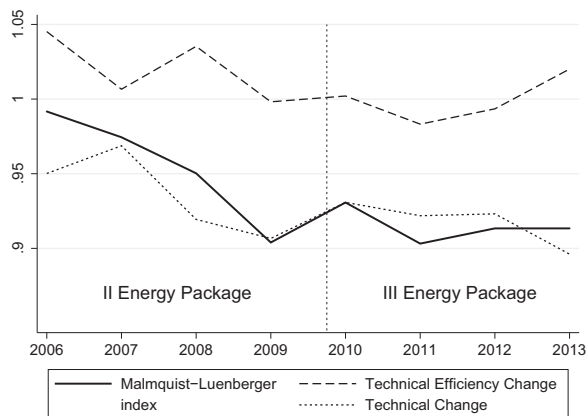


Fig. 4. Average Malmquist-Luenberger index.

capacity in use in the Benelux countries. Similarly, the capacities of the Nordic EU states are aggregated. Since 1996, the functioning of the energy firms in these Nordic counties has been subject to the common Nord Pool legislation.

Simar and Wilson (2007) argue that a two-stage DEA approach could be biased in the case of a high correlation between the first-stage input-output variables and the second-stage controls. The actual correlation matrix is reported in Table 4. Note that the absolute values of the correlations between the controls and the DEA components do not exceed 0.6. Thus, all of the included second-stage regressors are appropriate.

For each set of results, I compute the average partial effects (APEs) of one completed merger. These effects vary across the range of tested models. For the sake of simplicity, I exclusively report the APEs resulting from the model with included controls of vertical integration,

merger experience, management and finance. It is also worth to note that I define the short-run effects as the ones of the same-year mergers. The medium-run effects correspond to the ones of the mergers that have been completed two years ago.

3. Results and discussion

3.1. A first look at the eco-efficiency of the European energy sector

The eco-efficiency of European electricity producers over the 2005–2013 period ranges from 0.1832 to 1. The computed DEA eco-efficiency scores are reported in Table A.1 in the Appendix. The mean score was 0.7043, which suggests that, on average, electricity firms could increase their eco-efficiency by approximately 29.57%. The evolution of the entire sector's eco-efficiency is particularly interesting for policymakers. This trend can be captured by the yearly averages of the eco-efficiency scores, which are represented graphically in Fig. 3. It is worth comparing these to the average values of the MLPI (Fig. 4). Note that the trend predicted by the MLPI is quite similar, despite minor differences for 2011–2013. The same graph shows the decomposition of the MLPI into MLTEC and MLTC. Note that the sector's technical efficiency declines over the entire period of interest. This trend contradicts previously documented moderate efficiency gains that are specific to the early stages of the European energy liberalization.¹⁸

Between 2005 and 2009, the overall eco-efficiency of the sector, as well as its technical efficiency, decreased consistently. Meanwhile, allocative efficiency has changed ambiguously. This period was governed by the second electricity directive. It is worth comparing the eco-efficiency pattern to the patterns of mergers of different types (see Fig. 2). Horizontal cross-border mergers prevailed during this period. Meanwhile, the number of vertical cross-border mergers was steadily growing. Pollitt (2009) argues that vertical mergers reverse the unbundling policy implemented by the European Commission. Therefore, I hypothesize that vertical mergers have a negative impact on eco-efficiency.

The period from 2010 to 2013 was governed by the third EC electricity directive. Its adoption in September 2009 is a probable cause of the observed short-run eco-efficiency improvement. Note the sharp decrease in the number of vertical mergers, which was triggered by a more stringent policy of ownership unbundling. Allocative efficiency increases by the end of the period, which was mitigated by a downward shift of the sector's efficiency frontier. Thus, aggregate eco-efficiency over the 2010–2013 period has experienced marginal growth above the 2009 values.

The evolution of the eco-efficiency of separate utilities warrants investigation. I plot the computed individual scores in Fig. A.1.

¹⁸ See, e.g., Steiner (2001), Hattori et al. (2005) and Fiorio et al. (2007).

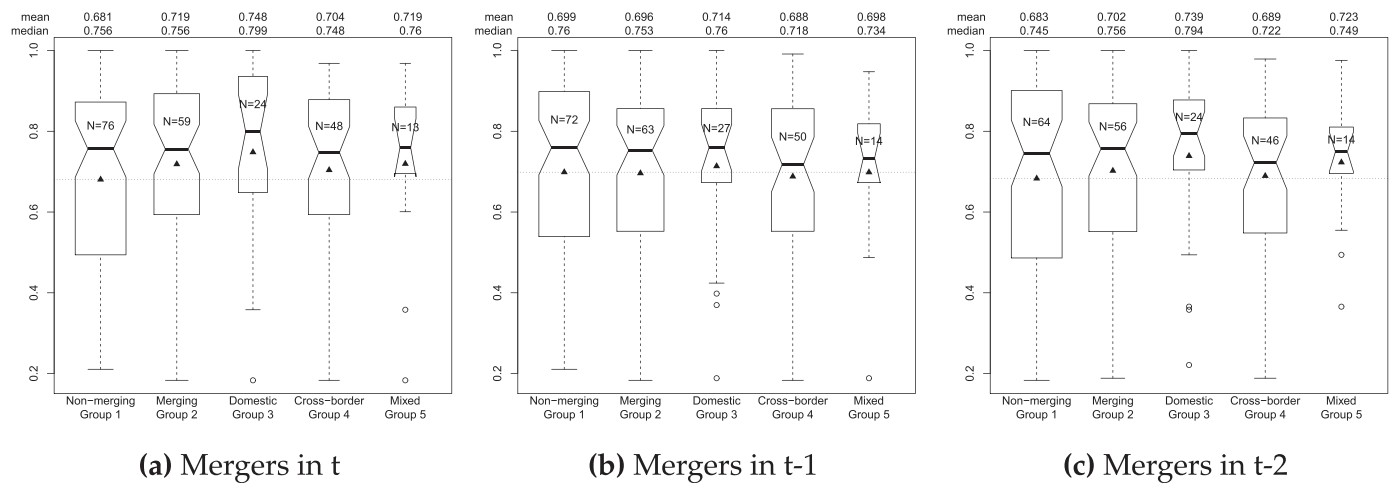


Fig. 5. Distribution of eco-efficiency scores with respect to completed domestic and cross-border M & As.

Table 5
Short-run impact of domestic and cross-border mergers on eco-efficiency.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CB mergers t	−0.05** (0.025)	−0.052*** (0.019)	−0.044** (0.019)	−0.076*** (0.026)	−0.067*** (0.02)	−0.058*** (0.014)	−0.052** (0.021)	−0.075*** (0.014)
Dom. mergers t	−0.061* (0.032)	−0.064 (0.04)	−0.089** (0.041)	−0.098*** (0.034)	−0.12*** (0.04)	−0.11** (0.041)	−0.1** (0.043)	−0.12*** (0.043)
Vertical integration			YES		YES	YES	YES	YES
Merger experience			YES			YES	YES	YES
Management				YES	YES	YES		YES
Finance				YES	YES		YES	YES
Location		YES						
Observations	135	135	135	135	135	135	135	135

Notes: Table reports estimations of Quasi maximum likelihood probit model. Dependent variable is DEA eco-efficiency score. All models contain year dummies for 2004 through 2013 and time averages of all explanatory variables. Standard errors are clustered on firm level. *, ** and *** indicate significance at the 10, 5% and 1% levels.

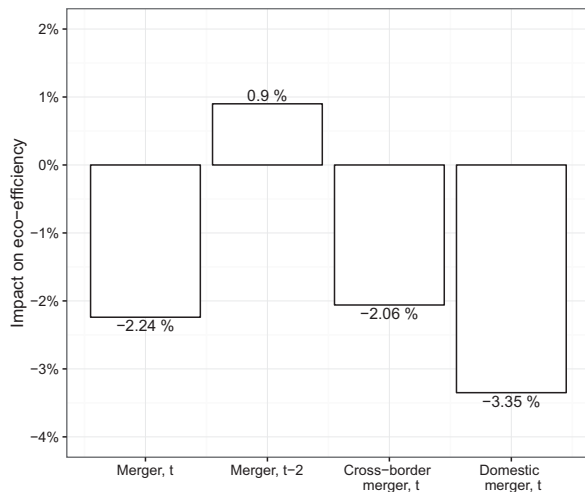


Fig. 6. Average partial effects of cross-border and domestic mergers on DEA eco-efficiency.

Électricité de France and Fortum were typically close to the best-practice frontier throughout the study period. The lowest average efficiency values were exhibited by Edison, Enel and Energias de Portugal. Overall, the eco-efficiency of individual firms is rather ambiguous. However, some patterns can be identified by classifying utilities in terms of the regulatory environment. National-level regula-

tion is expected to affect ecological, environmental and financial aspects of the functioning of energy utilities.

Perrin (2013) classifies the national regulatory models applied in Europe. Incentive-based regulatory models are in place in the Czech Republic, France and Germany (among other countries). The eco-efficiency scores of utilities with headquarters in these countries are illustrated in Fig. A.1a in the Appendix; they have demonstrate convergence to the interval [0.7,0.8]. I further distinguish firms with headquarters in Italy, Spain and Finland. These firms are exposed to the combined type of regulation, which is a mixture of the incentive-based and income-cap models. Greater dispersion in eco-efficiency scores is observed among these utilities. However, I find some convergence to the interval [0.3,0.5]. Thus, incentive-based regulation may favor eco-efficiency.

3.2. Distinguishing domestic and cross-border M & As

The distribution of eco-efficiency scores should guide preliminary predictions regarding the impacts of mergers. I divide all observations into groups based on the type of completed merger. For each group, I construct a box plot whose width is proportional to the number of observations. Each box plot has notches corresponding to the 95% confidence interval of the median. First, I compare the eco-efficiency of merging and non-merging utilities (group 1 and group 2 in Fig. 5). Non-merging utilities constitute the baseline group in further regression analyses. Mergers and non-mergers have almost identical median scores across the three periods of interest. The mean eco-efficiency of

Table 6
Short-run impact of domestic and cross-border mergers on Malmquist-Luenberger index.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CB mergers	−0.013* (0.0072)	−0.011*** (0.0033)	−0.014** (0.0053)	−0.017** (0.0071)	−0.015** (0.0052)	−0.015*** (0.0038)	−0.015** (0.0061)	−0.016*** (0.0041)
Dom. mergers	−0.0071 (0.0082)	−0.0084 (0.0095)	−0.018 (0.011)	−0.015* (0.0074)	−0.016 (0.0092)	−0.021** (0.0095)	−0.023* (0.012)	−0.025** (0.010)
Vertical integration			YES		YES	YES	YES	YES
Merger experience			YES			YES	YES	YES
Management				YES	YES	YES		YES
Finance				YES	YES			YES
Location		YES					YES	
Observations	120	120	120	120	120	120	120	120
R square	0.25	0.35	0.33	0.41	0.41	0.43	0.35	0.45

Notes: Table reports estimations of OLS model. Dependent variable is Malmquist-Luenberger productivity index. All models contain year dummies for 2006 through 2013 and firm fixed effects. Robust standard errors are clustered on firm level. *, ** and *** indicate significance at the 10, 5% and 1% levels.

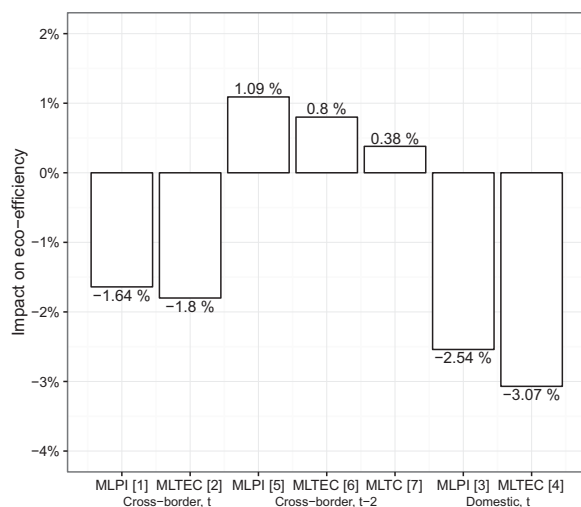


Fig. 7. Average partial effects of cross-border and domestic mergers on Malmquist-Luenberger index.

merged firms is marginally higher in the year the deal is completed.

The regression results suggest that any merger reduces the same-year eco-efficiency of electricity producers by an average of 2.24%. The corresponding coefficients are, in most cases, significant at the 1% level (Table A.3a in the Appendix). I find that a merger that was completed one year ago does not influence the eco-efficiency (results not

reported). Instead, such a merger exerts a positive but limited impact in the medium run. The estimated coefficients are reported in Table A.3b in the Appendix, and the corresponding average partial effect is 0.9%.

I further compare the eco-efficiency of electricity producers that have engaged in domestic and cross-border mergers. Approximately twice as many firms finalized international deals. The mean and median eco-efficiency values of firms that merged domestically are higher over all periods of interest (see Fig. 5). In the mixed type merger, both domestic and cross-border mergers are completed in the same year. Note that median eco-efficiency value in such cases is similar to the corresponding figure for non-mergers, whereas the mean eco-efficiency value is higher.

The results of the fractional regression model suggest that both cross-border and domestic mergers are detrimental to eco-efficiency in the year of their completion. The estimated coefficients are reported in Table 5. The impact of a finalized international merger is −2.06%. Each domestic merger decreases eco-efficiency by 3.35%, on average. Note that the overall short-run negative effect of mergers is confirmed. The magnitudes are compared in Fig. 6. I observe positive but non-significant impacts of both cross-border and domestic mergers over a two-year horizon (see Table A.4b in Appendix).

The estimation results of the short-run impact of mergers on the MLPI are reported in Table 6. The negative impacts of both cross-border and domestic mergers are confirmed. I also found statistically significant effect of cross-border mergers on the MLPI. I illustrate the average partial effects in Fig. 7. Note that the magnitudes of the short-run effect from the MLPI are smaller compared to those of the DEA

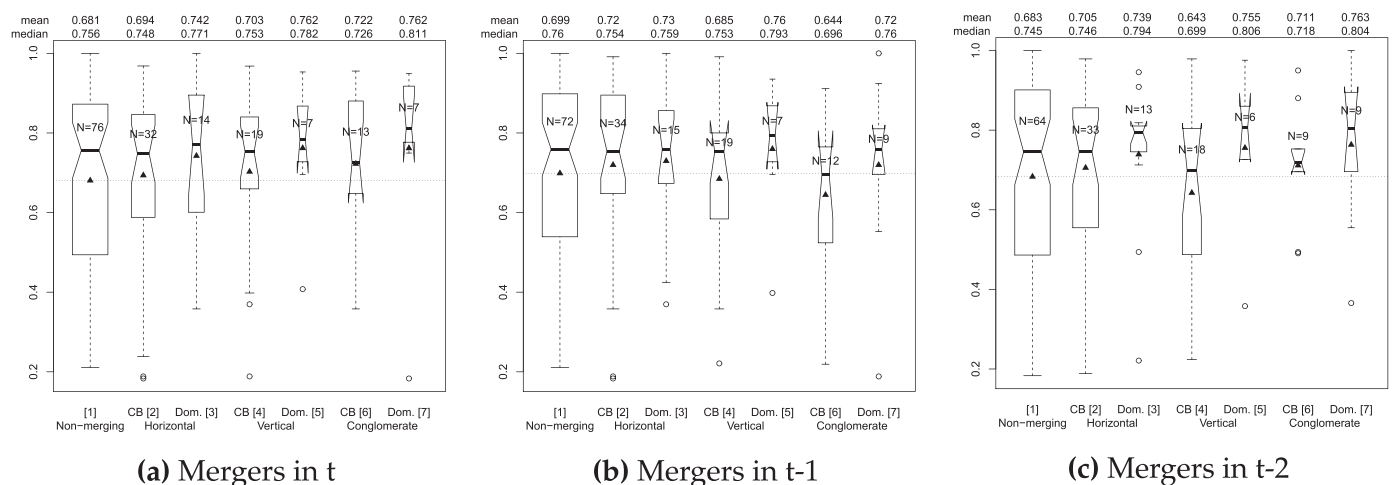


Fig. 8. Distribution of eco-efficiency scores with respect to completed Horizontal, Vertical, and Conglomerate M & As.

Table 7
Short-run impact of Horizontal, Vertical, and Conglomerate M & As on eco-efficiency.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Horizontal CB	−0.096** (0.043)	−0.089** (0.044)	−0.077* (0.045)	−0.12*** (0.036)	−0.11*** (0.03)	−0.078** (0.036)	−0.092*** (0.034)
Horizontal Dom.	−0.044 (0.049)	−0.1 (0.069)	−0.1 (0.071)	−0.05 (0.052)	−0.12* (0.061)	−0.084 (0.056)	−0.11 (0.082)
Vertical CB	0.0042 (0.035)	−0.028 (0.04)	−0.012 (0.035)	−0.049 (0.052)	−0.053 (0.049)	−0.031 (0.043)	−0.011 (0.039)
Vertical Dom.	−0.14* (0.072)	−0.14* (0.071)	−0.12 (0.092)	−0.15*** (0.05)	−0.13** (0.053)	−0.13*** (0.05)	−0.12 (0.074)
Conglomerate CB	0.043 (0.054)	0.12* (0.069)	0.15* (0.078)	0.035 (0.063)	0.11* (0.066)	0.12** (0.058)	0.19** (0.086)
Conglomerate Dom.	−0.014 (0.1)	−0.066 (0.088)	−0.058 (0.081)	−0.12 (0.11)	−0.2* (0.11)	−0.18** (0.072)	−0.082 (0.11)
Vertical integration		YES	YES		YES	YES	YES
Merger experience			YES			YES	YES
Management				YES	YES	YES	
Finance				YES	YES		YES
Observations	135	135	135	135	135	135	135

Notes: Table reports estimations of Quasi maximum likelihood probit model. Dependent variable is DEA eco-efficiency score. All models contain year dummies for 2004 through 2013 and time averages of all explanatory variables. Standard errors are clustered on firm level. *, ** and *** indicate significance at the 10, 5% and 1% levels.

Table 8
Medium-run impact of Horizontal, Vertical, and Conglomerate M & As on eco-efficiency.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Horizontal CB t-2	0.062*** (0.024)	0.074*** (0.027)	0.079*** (0.029)	0.045 (0.032)	0.054** (0.025)	0.049* (0.029)	0.083*** (0.024)
Horizontal Dom. t-2	0.02 (0.054)	0.03 (0.046)	−0.0068 (0.051)	0.026 (0.047)	0.034 (0.039)	−0.023 (0.048)	0.0066 (0.043)
Vertical CB t-2	−0.11*** (0.037)	−0.11*** (0.037)	−0.1*** (0.035)	−0.099** (0.04)	−0.075** (0.036)	−0.081*** (0.026)	−0.077** (0.033)
Vertical Dom. t-2	−0.057 (0.082)	−0.059 (0.081)	0.0029 (0.074)	−0.072 (0.08)	−0.073 (0.075)	0.00021 (0.076)	−0.027 (0.069)
Conglomerate CB t-2	0.019 (0.052)	0.054 (0.05)	0.081 (0.059)	0.044 (0.046)	0.072 (0.057)	0.079 (0.074)	0.14** (0.065)
Conglomerate Dom. t-2	0.1 (0.068)	0.13* (0.068)	0.15** (0.073)	0.13** (0.062)	0.15* (0.076)	0.16** (0.065)	0.23*** (0.075)
Vertical integration		YES	YES		YES	YES	YES
Merger experience			YES			YES	YES
Management				YES	YES	YES	
Finance				YES	YES		YES
Observations	120	120	120	120	120	120	120

Notes: Table reports estimations of Quasi maximum likelihood probit model. Dependent variable is DEA eco-efficiency score. All models contain year dummies for 2004 through 2013 and time averages of all explanatory variables. Standard errors are clustered on firm level. *, ** and *** indicate significance at the 10, 5% and 1% levels.

eco-efficiency. The decomposition of the MPLI reveals that in the short run, the aggregate effect is mainly attributed to technical efficiency changes (results not reported). However, the position of the best-practice frontier captured by technical changes (i.e., the MLTC) is influenced by cross-border merger activity in the medium run.

The short-run negative impact of mergers is in line with the results established in the broad and heterogeneous general M & A literature. Firm-level efficiency changes have been systematically proxied by alterations in shareholder wealth. The cross-industry empirical results suggest that, at best, mergers do not impact the acquirers (e.g.,

Asquith, 1983; Agrawal et al., 1992; Loderer and Martin, 1992), but they are often detrimental (e.g., King et al., 2004; Moeller et al., 2004).

A relevant theoretical explanation is provided by the concept of "lemons" for sale, whereby poorly performing firms are targeted for M & As (e.g., Jensen and Ruback, 1983; Lichtenberg et al., 1987). The recent theoretical models of Neary (2007) and Nocke and Yeaple (2007) support this hypothesis. McGuckin and Nguyen (1995) argued that large, inefficient plants are more likely to be sold than closed. Balsvik and Haller (2010) revealed that decrease in total factor productivity usually precedes acquisition. Blonigen et al. (2014)

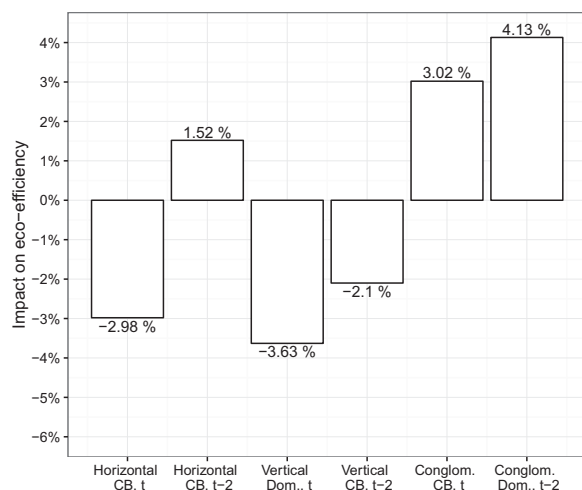


Fig. 9. Average partial effects of mergers on DEA eco-efficiency.

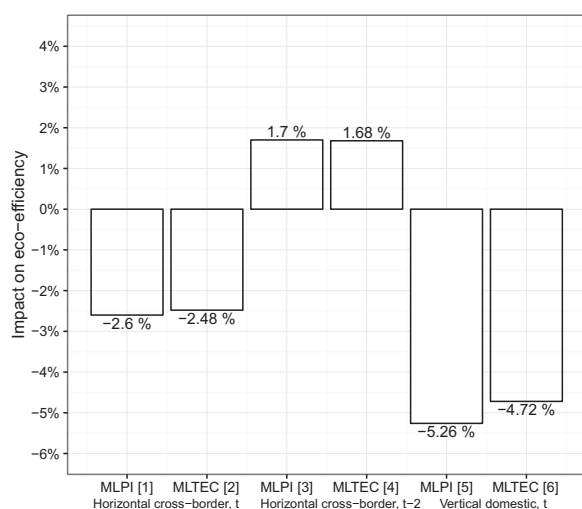


Fig. 10. Average partial effects of mergers of different types on Malmquist-Luenberger index.

reported that recent negative productivity shocks significantly increase the likelihood that a firm will be acquired by a foreign multinational.

Berry (2000) provided the following interpretation of the negative outcomes of domestic electric utility mergers. Domestic acquirers are less uncertain about the target's future performance than bidders in international mergers. Domestic acquirers are more familiar with the national market and legislation and are better able to predict risks. They are more likely to pay a deal premium, which could be in terms of shareholders wealth or firm efficiency. Therefore, domestic acquirers tend to absorb less-efficient targets first.

Cross-border M & As remain relatively understudied compared to domestic mergers. Modern theory is ambiguous regarding their effects on efficiency. On the one hand, international M & As should be beneficial to firm-level efficiency because of newfound access to the country-specific capabilities of a foreign target. Nocke and Yeaple (2007) theoretically demonstrated that M & As are caused by heterogeneity in firms' intangible assets. Cross-border mergers are motivated by complementarities between internationally mobile and non-mobile capabilities.¹⁹ On the other hand, a number of theoretical channels

suggest that cross-border mergers could hamper a firm's performance (Bertrand and Betschinger, 2012). First, acquirers could experience greater informational asymmetry: They could overvalue the potential gains from synergy and, consequently, overpay for target shares. Second, international bidders face the costs of adapting to the macro-economic, legislative and cultural conditions of the destination country. Finally, acquirers must overcome the increased complexity of coordinating with foreign affiliates.

The result that cross-border mergers are detrimental to acquirers is in line with empirical M & A studies that have found negative effects on bidders' wealth.²⁰ However, researchers in this area are far from reaching a consensus. An important body of literature documents positive wealth effects of cross-border acquisitions for bidders.²¹ Bertrand and Betschinger (2012) found that both domestic and cross-border M & As negatively influence the financial efficiency of acquirers. They argue that cross-border mergers can have weaker anti-competitive effects. Because partners in international deals are more remote, their mutual pre-merger competition is lower, which implies lower efficiency alterations following cross-border mergers.

3.3. Distinguishing among horizontal, vertical, and conglomerate M & As

The next step in this study is to differentiate among the effects of horizontal, vertical and conglomerate mergers. The distributions of eco-efficiency scores (see Fig. 8) indicate that horizontal and conglomerate mergers should positively contribute to eco-efficiency. Vertical deals should have positive outcomes over the short run. In the medium run, vertical cross-border mergers should cause a decline in eco-efficiency, whereas vertical domestic ones should have a positive impact.

The regression results suggest that horizontal domestic mergers do not have any statistically significant effects on eco-efficiency (see Tables 7, 8). Therefore, I conclude that electricity producers do not obtain the expected synergy gains from domestic mergers. It is worth noting that current European Merger regulation primarily concerns domestic horizontal mergers.²² Since I do not find any negative effects of such deals, I conclude that the European Commission successfully protects the electricity industry from some eco-efficiency losses.

The results suggest that horizontal cross-border mergers damage eco-efficiency over the short run, and the corresponding average partial effect is -2.98%. The corresponding magnitude for the MLPI is -2.6% (compare Figs. 9–10). It is worth noting that a merger of this type is more detrimental to eco-efficiency than an average merger (-2.24%). Horizontal cross-border mergers start to positively affect eco-efficiency in the medium run. The average partial effect of such a merger is 1.52% (1.7% for MLPI). This result is similar to the findings of Becher et al. (2012), who report a gain of 1.75% in the wealth of combined firms. Datta et al. (2013) study the long-run effects of EU energy mergers over the 1990–2006 period, and they come to the opposite conclusion, revealing losses of 6.6% in the shareholder wealth of combined utilities. Becker-Blease et al. (2008) find a similar negative effect (-6.77%) for 3-year buy-and-hold returns.

(footnote continued)

to absorb rents from market inefficiency. Spillovers from corporate governance standards could improve corporate governance (Martynova and Renneboog, 2008). Shimizu et al. (2004) found that national cultural differences and organizational learning induce post-merger value creation. Kiyamaz and Mukherjee (2000) argued that an acquirer is better able to make use of its strategic advantages when it engages in cross-border acquisitions. Targets can benefit from the optimization of their business processes and novel access to cheaper external resources.

²⁰ See, e.g., Eckbo and Thorburn (2000), Denis et al. (2002), Aw and Chatterjee (2004), Moeller et al. (2005), Aybar and Ficici (2009), Chen and Young (2010).

²¹ See, e.g., Markides and Ittner (1994), Akhigbe and Martin (2000), Boateng et al. (2008), Francis et al. (2008), Chari et al. (2010), Gubbi et al. (2010), Bhagat et al. (2011).

²² See European Commission (2004) and European Commission (2008)

¹⁹ Other studies have listed specific benefits of international mergers. Servaes and Zenner (1994) argued that such deals allow firms to exploit differences in tax systems and

The previous M & A literature devoted to the electricity industry has generally identified negative short-run effects of domestic horizontal mergers but has not captured any effects of cross-border horizontal deals. Leggio and Lien (2000) found that US acquirers experienced a short-term wealth loss of 0.79% around the announcement of electric mergers over 1994–1996. Datta et al. (2013) confirmed short-run losses of 0.1% in the shareholder wealth of combined utilities. Similarly, Becher et al. (2012) addressed horizontal mergers in the US electricity market in 1993–2004, and they revealed short-run losses of 1.36% for acquirers. The two papers are outliers. Berry (2000) finds positive short-run cumulative abnormal returns for acquirers' shareholders of 0.9% but finds no statistically significant excess returns for the shareholders of acquirers in cross-border mergers. Becker-Blease et al. (2008) addresses post-merger alterations in terms of both the wealth and the financial performance of merged US energy utilities. They documented a post-merger loss of 8.7% in net present value.

A delayed-in-time positive effect can be interpreted based on the idea that acquired underperforming lemons later evolve into overperforming “cherries”. This reasoning is based on the dynamic model of cross-border M & A activity elaborated by Blonigen et al. (2014). They argue that multinationals aim to acquire promising assets immediately after a negative productivity shock because their price is relatively low at that point. Once the integration of an underperforming energy utility (a lemon) in the production structure is finalized, and it becomes an overperforming cherry. The two-year delay can be explained in terms of the time needed to perform internal post-merger adjustments. The studied firms attempt to adapt their managerial practices and administrative structures to fully integrate a new entity. A plausible industry-specific interpretation is that a certain amount of time is required to reshape the energy mix utilized for electricity production.

Despite the evidence that vertical mergers represent a significant mode of foreign direct investment (e.g., Carr et al., 2001), they still have not received proper attention in the M & A literature. Alfaro and Charlton (2009) were the first to address vertical and horizontal mergers at the deal level and developed an approach to differentiating between these types of mergers. The previous M & A literature reveals no impact of diversifying energy mergers (Leggio and Lien, 2000; Becker-Blease et al., 2008). Becher et al. (2012) found that the effect of a non-horizontal utility deal is similar in magnitude to that of a horizontal deal, although this effect is not significant. Bednarczyk et al. (2010) argued that horizontal energy acquisitions induce stronger positive market reactions than do vertical or diversifying deals.

The regression results suggest that, in general, vertical mergers negatively impact the eco-efficiency of European electricity producers. Vertical domestic mergers are detrimental in the short run, with an average partial effect of −3.63% (see Fig. 9), but in the medium run, the effect is no longer statistically significant. The corresponding effect for the MLPI is −5.26% (see Fig. 10). The opposite pattern is observed for international mergers, which do not affect eco-efficiency in the year of their finalization. The average partial effect of a cross-border vertical merger two years after its completion is −2.1% (see Table 8). Thus, domestic and cross-border vertical mergers differ in the timing of their effects.

Firms that engage in vertical mergers expect gains arising from supply chain coordination. However, recent studies sharply conflict with this view. Atalay et al. (2014) argues that vertical ownership stimulates efficient intra-firm transfers of organizational capabilities and other intangible inputs rather than physical shipments alone. Ramondo et al. (2016) revealed that vertical relatedness does not determine intra-firm trade in goods. Alfaro et al. (2016) argued that the administrative cost of integration is the limiting factor for productivity gains from vertical integration. They further proved empirically that price-takers can compensate for excessive administrative expenditures exclusively through higher output prices. Thus, the revealed negative impact of vertical mergers may be attributable to important integration costs.

The theory predicting gains from conglomerate M & As dates to the 1970s. Lewellen (1971) argued that conglomerate acquisitions could generate financial synergy due to co-insurance effects. If the cash flows of merging entities are imperfectly correlated, their combined volatility will decrease, leading to a lower default risk for the consolidated company. However, empirical evidence of this effect remains scarce. Breinlich (2008) found that conglomerate mergers constitute a significant part of total M & As activities. He underlined two principal motives behind acquiring assets in unrelated industries: access to new export markets and cheaper assets because of high import competition. Herger and McCorriston (2016) highlighted the empirical importance of horizontal and vertical FDI in a sample of cross-border acquisitions for the 1990–2011 period. In addition, they discovered that more than 20% of all M & As are conglomerate mergers. Furthermore, conglomerate mergers were proven to induce waves of international M & As.

My regression results suggest that conglomerate mergers positively impact the DEA eco-efficiency. The short-run effect of a cross-border merger of this type is 3.02% (see Fig. 9). Over a two-year horizon, domestic conglomerate mergers increase eco-efficiency by 4.13%. Thus, the effects of conglomerate mergers are the opposite of those produced by vertical mergers. However, the regression results for the MLPI (see Table A.5 and Table A.5 in the Appendix) do not confirm the statistical significance of conglomerate mergers. Thus, there is limited empirical evidence regarding conglomerate mergers.

4. Conclusions and policy implications

Since the 1990s, the European electricity industry has undergone extensive liberalization. Market players reacted by engaging in waves of M & As. This paper addresses how this surge of M & As influenced the eco-efficiency of energy producers in the short run (same year) and medium run (two years later). This is the first attempt (to the best of my knowledge) to quantify the eco-efficiency outcomes of the late stages of EU electricity market reforms. The results suggest that the average eco-efficiency of the European energy sector decreased over the 2005–2009 period, and this trend accelerated in 2008, possibly because of the global financial crisis. For 2009–2013, eco-efficiency was, on average, at the 2009 level. This trend contradicts previously documented moderate efficiency gains that are specific to the early stages of European energy liberalization. The pattern of individual eco-efficiency scores suggests that incentive-based regulation of electricity distribution should be promoted.

This study demonstrates that a merger of any type has a detrimental effect on eco-efficiency of 2.24% in the short run and a positive effect of 0.9% in the medium run. I also find that the domestic deals are systematically more detrimental than the cross-border deals. Therefore, the regulators should treat cross-border and domestic mergers in different ways. It is appropriate to predict the outcomes of potential mergers using ex-ante Data envelopment analysis, although such an exercise might be an issue for particular national regulatory bodies. Regulators should create a common cross-country comprehensive dataset on the various dimensions of electricity production.²³ Based on such a dataset, regulators would be able to make decisions that are consistent with common ecological objectives.

Since this study is dedicated to the European electricity sector, it is pertinent to discuss the implications for EU merger regulation. According to the regulations in place, the European Commission builds its decisions regarding potential mergers primarily based on outcomes for competition. The only way to account for post-merger efficiency alterations is related to efficiency claims.²⁴ Despite the common EU

²³ Jamasb and Pollitt (2003) has previously proposed the coordination of international benchmarking exercises.

²⁴ See Section VII “Efficiencies” in European Commission (2004). In order to support potential mergers, firms should justify that any resulting efficiency is simultaneously beneficial to consumers, merger-specific and verifiable.

ecological objectives, current European merger regulation does not take into account the ecological dimension. Furthermore, efficiency claims are rarely emphasized by merging firms, which reduces their importance in regulator's decisions. I suggest that the practice of efficiency claims should be withdrawn and replaced with an indispensable eco-efficiency analysis of potential mergers. To this end, the European Commission should collaborate with ACER.

This paper represents one of the first attempts to distinguish between the effects of horizontal and non-horizontal mergers in the electricity industry. Horizontal domestic mergers do not affect eco-efficiency. This result should be attributed to the close scrutiny of the European Commission of horizontal domestic deals. On the one hand, a harmful impact is avoided. On the other hand, efficiency is not promoted. Therefore, this result confirms the need to ground merger regulation in efficiency estimates.

I find that horizontal cross-border mergers have negative short-run effects of approximately 3%. However, over the medium run, deals of this type increase eco-efficiency by roughly 1.5%. A plausible interpretation is that initially underperforming targets (lemons) evolve into overperforming entities (cherries) after being acquired. Thus, I would suggest that the electricity firms consider international intra-industry expansion as a priority strategy. An associated policy suggestion is to systematically address the efficiency outcomes arising at different points of time after the completion of a merger.

I found evidence, albeit limited, that conglomerate mergers have positive effects on eco-efficiency. European electricity producers should

consider merging with unrelated businesses because this strategy is at least not harmful. Meanwhile, A vertical domestic merger tends to reduce eco-efficiency by approximately 3.6% in the year of its completion. Over the medium term, cross-border vertical mergers reduce eco-efficiency by 2.1%. Current EU merger regulation is not concerned with vertical mergers.²⁵ I suggest that the vertical mergers deserve the special attention of regulators, especially in industries that are responsible for important shares of emissions.

It is worth emphasizing directions of further research. First, the effects of mergers on the eco-efficiency of small electricity firms should be studied. Second, the application of non-radial slack-based models should be considered. Finally, attention should be paid to differentiating among mergers of different sizes.

Acknowledgements

The author appreciates the valuable comments of two anonymous referees. The author would like to acknowledge Lionel Fontagné, Katrin Millock, Sandra Poncet and Farid Toubal for their insights into and discussions of early versions of the paper. All remaining errors are my own. Special thanks are owed to David Thesmar for providing access to the SDC Platinum, Thomson One and BvD Orbis databases. The author is particularly grateful to LABEX OSE, the Paris School of Economics and Paris 1 Panthéon Sorbonne University for funding access to the Enerdata datasets.

Appendix

(Table A.4a) (Table A.6) (Table A.7).

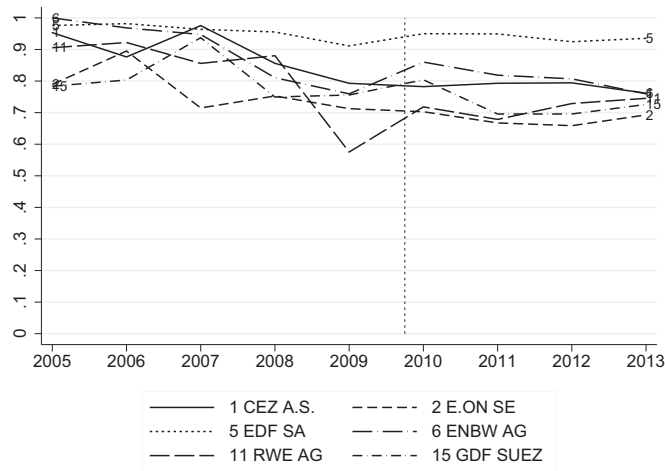
Table A.1

DEA eco-efficiency scores.

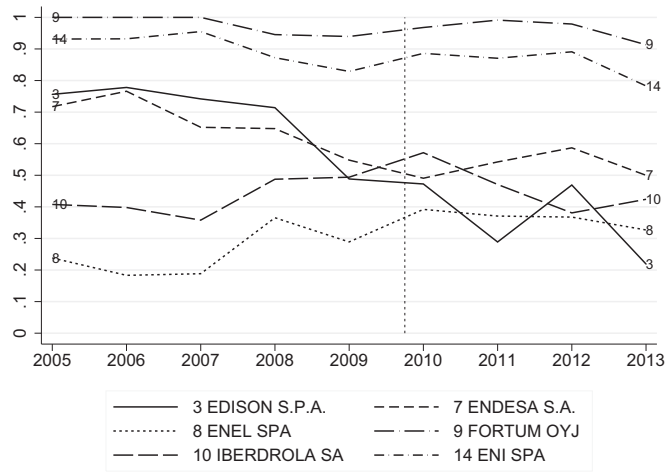
Firm name	Headquarter location	2005	2006	2007	2008	2009	2010	2011	2012	2013	mean
CEZ A.S.	Prague, Czech Republic	0.9533	0.8764	0.9757	0.8561	0.7929	0.7824	0.793	0.7944	0.7616	0.8429
E.ON SE	Dusseldorf, Germany	0.7895	0.8951	0.7149	0.7528	0.7126	0.7029	0.6675	0.6587	0.6928	0.7319
EDISON S.P.A.	Milan, Italy	0.7562	0.7781	0.7419	0.714	0.4884	0.4725	0.2888	0.4692	0.219	0.5476
EDP S.A.	Lisbon, Portugal	0.5524	0.5549	0.3696	0.2211	0.2237	0.2785	0.2353	0.2105	0.2498	0.3218
EDF SA	Paris, France	0.976	0.9819	0.9638	0.9555	0.9113	0.95	0.9493	0.9246	0.9353	0.9497
ENBW AG	Karlsruhe, Germany	1	0.9683	0.9475	0.8105	0.7595	0.8601	0.8186	0.807	0.7596	0.859
ENDESA S.A.	Madrid, Spain	0.7173	0.7661	0.652	0.6478	0.5481	0.4907	0.5423	0.5869	0.5	0.6057
ENEL SPA	Rome, Italy	0.2382	0.1832	0.1884	0.3656	0.289	0.392	0.3709	0.3676	0.3269	0.3024
FORTUM OYJ	Espoo, Finland	0.9314	0.9321	0.9551	0.8724	0.8289	0.886	0.8706	0.891	0.783	0.8834
IBERDROLA SA	Bilbao, Spain	1	1	1	0.9455	0.9395	0.968	0.9914	0.979	0.9134	0.9708
RWE AG	Essen, Germany	0.7837	0.8033	0.9372	0.7495	0.7559	0.804	0.6956	0.6958	0.726	0.7723
VATTENFALL AB	Stockholm, Sweden	0.4077	0.398	0.358	0.4875	0.494	0.5715	0.4709	0.3809	0.424	0.4436
VERBUND AG	Vienna, Austria	0.9056	0.922	0.8559	0.8803	0.5753	0.7182	0.6786	0.7288	0.7454	0.7789
ENI SPA	Rome, Italy	1	0.9088	0.9273	0.8911	0.7599	0.8269	0.833	0.9134	0.8877	0.8831
GDF SUEZ	Courbevoie, France	0.79	0.6042	0.6008	0.6734	0.7456	0.6873	0.5362	0.6066	0.7962	0.6711

Notes: Table reports calculated DEA eco-efficiency scores. A firm has the unity score in a given year if in this year it is most efficient comparing to all other utilities over 9-year period.

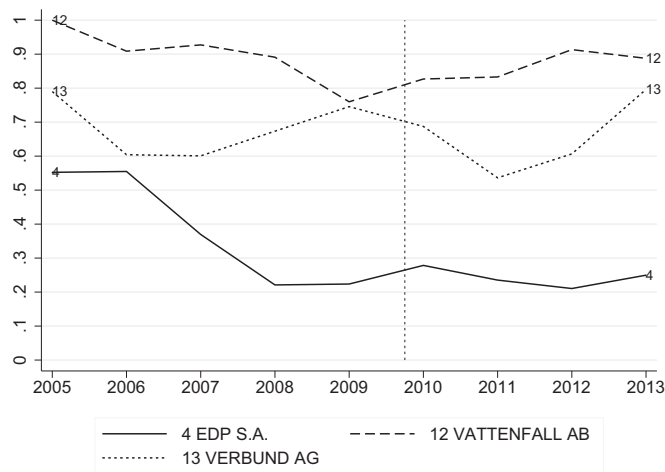
²⁵ See paragraph 25 European Commission (2008).



(a) Incentive-based regulatory model



(b) Incentive and revenue cap combined regulatory model



(c) Other types of regulation

Fig. A.1. Individual eco-efficiency scores.

Table A.2

Repartition of targets by sectors.

U.S. SIC	Sector of activity	Relatedness	Domestic	Cross-border
1221	Bituminous coal and lignite surface mining	vertical	2	0
1311	Crude petroleum and natural gas	vertical	0	8
1522	General contractors-residential buildings other than single-family	conglomerate	0	1
1731	Electrical work	conglomerate	1	0
2813	Industrial gases	conglomerate	0	1
3433	Heating equipment except electric and warm air furnaces	conglomerate	1	0
3511	Steam gas and hydraulic turbines and turbine generator set units	vertical	0	3
3621	Motors and generators	vertical	0	1
3674	Semiconductors and related devices	conglomerate	1	0
4911	Electric services	horizontal	16	52
4922	Natural gas transmission	conglomerate	4	2
4923	Natural gas transmission and distribution	conglomerate	1	1
4924	Natural gas distribution	conglomerate	0	2
4931	Electric and other services combined	horizontal	1	0
4939	Combination utilities not elsewhere classified	horizontal	0	1
4941	Water supply	conglomerate	1	0
499A	Cogeneration, alternative energy sources	vertical	7	14
5172	Petroleum and petroleum products wholesalers except bulk stations and terminals	conglomerate	0	1
6512	Operators of nonresidential buildings	conglomerate	1	0
6719	Offices of holding companies not elsewhere classified	conglomerate	0	1
7389	Business services not elsewhere classified	conglomerate	0	1
8711	Engineering services	conglomerate	0	2
8742	Management consulting services	conglomerate	0	1
8748	Business consulting services not elsewhere classified	conglomerate	0	1

Notes: Table reports primary sectors of target firms' activities.

Table A.3a

Short-run impact of any-type mergers on eco-efficiency.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mergers t	−0.052** (0.022)	−0.052*** (0.02)	−0.05*** (0.016)	−0.081*** (0.023)	−0.08*** (0.017)	−0.062*** (0.015)	−0.064*** (0.02)	−0.081*** (0.015)
GenToSales			0.87** (0.38)		0.76** (0.34)	0.8*** (0.3)	0.94** (0.4)	0.77** (0.32)
TwoMergers			−0.067 (0.066)			−0.029 (0.057)	−0.086 (0.085)	−0.034 (0.061)
TwoCBMergers			0.032 (0.062)			−0.011 (0.065)	0.057 (0.076)	−0.0082 (0.077)
K/S				0.47*** (0.096)	0.31*** (0.051)	0.25*** (0.054)		0.31*** (0.046)
I/K				−1.3*** (0.42)	−1.7*** (0.38)	−1.3*** (0.4)		−1.6*** (0.38)
Y/S				−2.4*** (0.54)	−2.7*** (0.48)	−2.7*** (0.47)		−2.6*** (0.43)
Leverage				−0.02 (0.06)	−0.044 (0.07)		−0.095 (0.098)	−0.066 (0.067)
Leverage square				0.0017 (0.0061)	0.0039 (0.0074)		0.012 (0.0094)	0.0058 (0.007)
CurrentRatio				0.91* (0.5)	0.8* (0.47)		0.45 (0.58)	0.74* (0.41)
CurrentRatio square				−0.38** (0.16)	−0.34** (0.15)		−0.2 (0.2)	−0.32** (0.14)
Observations	135	135	135	135	135	135	135	135

Notes: Table reports estimations of Quasi maximum likelihood probit model. Dependent variable is DEA eco-efficiency score. All models contain year dummies for 2004 through 2013 and time averages of all explanatory variables. Standard errors are clustered on firm level. *, ** and *** indicate significance at the 10, 5% and 1% levels.

Table A.3b

Medium-run impact of any-type mergers on eco-efficiency.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mergers t–2	0.014 (0.015)	0.018 (0.019)	0.028** (0.014)	0.0077 (0.02)	0.022 (0.015)	0.022 (0.015)	0.041** (0.017)	0.032** (0.016)
GenToSales			0.86** (0.38)		0.9** (0.38)	0.73** (0.32)	0.94** (0.39)	0.66** (0.33)
TwoMergers			–0.08 (0.092)			–0.095 (0.073)	–0.082 (0.12)	–0.12 (0.082)
TwoCBMergers			–0.11 (0.083)			–0.12 (0.081)	–0.14 (0.11)	–0.15 (0.1)
K/S				0.32** (0.15)	0.12 (0.11)	0.2** (0.096)		0.25*** (0.066)
I/K				–1.1 (0.68)	–1.3* (0.67)	–0.68 (0.47)		–0.64 (0.53)
Y/S				–2.1** (1)	–2.2** (0.89)	–2.2*** (0.65)		–2.1*** (0.76)
Leverage				0.00051 (0.071)	–0.0079 (0.086)		–0.12* (0.067)	–0.14** (0.072)
Leverage square				–0.0036 (0.0084)	–0.0036 (0.0093)		0.0071 (0.0067)	0.0083 (0.0081)
CurrentRatio				0.67 (0.64)	0.65 (0.58)		0.024 (0.57)	0.11 (0.54)
CurrentRatio square				–0.28 (0.21)	–0.27 (0.19)		–0.0096 (0.2)	–0.049 (0.19)
Observations	120	120	120	120	120	120	120	120

Notes: Table reports estimations of Quasi maximum likelihood probit model. Dependent variable is DEA eco-efficiency score. All models contain year dummies for 2004 through 2013 and time averages of all explanatory variables. Standard errors are clustered on firm level. *, ** and *** indicate significance at the 10, 5% and 1% levels.

Table A.4a

Short-run impact of domestic and cross-border mergers on eco-efficiency.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CB mergers t	–0.05** (0.025)	–0.052*** (0.019)	–0.044** (0.019)	–0.076*** (0.026)	–0.067*** (0.02)	–0.058*** (0.014)	–0.052** (0.021)	–0.075*** (0.014)
Dom. mergers t	–0.061* (0.032)	–0.064 (0.04)	–0.089** (0.041)	–0.098*** (0.034)	–0.12*** (0.04)	–0.11** (0.041)	–0.1** (0.043)	–0.12*** (0.043)
GenToSales			0.95** (0.38)		0.84** (0.34)	0.93*** (0.3)	0.95** (0.39)	0.83*** (0.31)
TwoMergers			–0.058 (0.087)			–0.016 (0.081)	–0.062 (0.098)	–5.3e–05 (0.077)
TwoCBMergers			0.0037 (0.072)			–0.051 (0.073)	0.025 (0.081)	–0.033 (0.083)
K/S				0.47*** (0.095)	0.28*** (0.046)	0.23*** (0.058)		0.3*** (0.043)
I/K				–1.2*** (0.43)	–1.7*** (0.38)	–1.3*** (0.42)		–1.5*** (0.35)
Y/S				–2.5*** (0.59)	–2.8*** (0.53)	–2.8*** (0.36)		–2.7*** (0.4)
Leverage				–0.037 (0.062)	–0.053 (0.065)		–0.1 (0.095)	–0.085 (0.063)

(continued on next page)

Table A.4a (continued)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Leverage square				0.0037 (0.0066)	0.0046 (0.0069)		0.012 (0.0087)	0.0077 (0.0068)
CurrentRatio				0.89* (0.5)	0.82* (0.47)		0.5 (0.56)	0.82** (0.4)
CurrentRatio square				−0.38** (0.16)	−0.35** (0.15)		−0.22 (0.2)	−0.36*** (0.13)
Observations	135	135	135	135	135	135	135	135

Notes: Table reports estimations of Quasi maximum likelihood probit model. Dependent variable is DEA eco-efficiency score. All models contain year dummies for 2004 through 2013 and time averages of all explanatory variables. Standard errors are clustered on firm level. *, ** and *** indicate significance at the 10, 5% and 1% levels.

Table A.4b

Medium-run impact of domestic and cross-border mergers on eco-efficiency.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CB mergers t−2	0.013 (0.019)	0.013 (0.023)	0.023 (0.022)	0.0084 (0.029)	0.024 (0.025)	0.013 (0.023)	0.043* (0.026)	0.029 (0.026)
Dom. mergers t−2	0.014 (0.021)	0.03 (0.032)	0.027 (0.031)	0.004 (0.022)	0.014 (0.027)	0.027 (0.031)	0.028 (0.036)	0.032 (0.037)
GenToSales			0.89** (0.39)		0.91** (0.38)	0.78** (0.34)	0.94** (0.39)	0.68** (0.34)
TwoMergers			−0.078 (0.099)			−0.1 (0.071)	−0.082 (0.12)	−0.12 (0.085)
TwoCBMergers			−0.13 (0.089)			−0.14* (0.079)	−0.15 (0.11)	−0.15 (0.1)
K/S				0.32** (0.15)	0.12 (0.11)	0.21** (0.098)		0.25*** (0.071)
I/K				−1 (0.69)	−1.3* (0.67)	−0.68 (0.51)		−0.5 (0.47)
Y/S				−2.1** (1)	−2.2** (0.9)	−2.2*** (0.62)		−2.2*** (0.73)
Leverage				−0.016 (0.073)	−0.014 (0.091)		−0.14* (0.084)	−0.15* (0.078)
Leverage square				−0.0019 (0.0082)	−0.0029 (0.0095)		0.0093 (0.0078)	0.0093 (0.0082)
CurrentRatio				0.65 (0.64)	0.64 (0.57)		−0.0058 (0.58)	0.16 (0.53)
CurrentRatio square				−0.28 (0.21)	−0.27 (0.19)		0.0015 (0.2)	−0.07 (0.19)
Observations	120	120	120	120	120	120	120	120

Notes: Table reports estimations of Quasi maximum likelihood probit model. Dependent variable is DEA eco-efficiency score. All models contain year dummies for 2004 through 2013 and time averages of all explanatory variables. Standard errors are clustered on firm level. *, ** and *** indicate significance at the 10, 5% and 1% levels.

Table A.5
Short-run impact of Horizontal, Vertical, and Conglomerate M & As on Malmquist-Luenberger index.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Horizontal CB t	−0.027*** (0.0073)	−0.023*** (0.0047)	−0.029*** (0.0060)	−0.025*** (0.0074)	−0.025*** (0.0057)	−0.024*** (0.0058)	−0.030*** (0.0071)	−0.026*** (0.0059)
Horizontal Dom. t	−0.0015 (0.014)	−0.011 (0.018)	−0.0079 (0.015)	−0.0058 (0.013)	0.00049 (0.012)	−0.0072 (0.012)	−0.012 (0.016)	−0.011 (0.012)
Vertical CB t	0.0023 (0.011)	0.0026 (0.010)	−0.00030 (0.0079)	−0.0091 (0.013)	−0.0071 (0.012)	−0.0082 (0.0077)	−0.00099 (0.0083)	−0.0085 (0.0081)
Vertical Dom. t	−0.027*** (0.0089)	−0.018 (0.014)	−0.054*** (0.014)	−0.029*** (0.0073)	−0.045*** (0.015)	−0.051** (0.017)	−0.056*** (0.016)	−0.053*** (0.017)
Conglomerate CB t	0.0082 (0.017)	0.0051 (0.018)	0.031 (0.020)	0.000018 (0.017)	0.0092 (0.018)	0.024 (0.018)	0.026 (0.020)	0.019 (0.018)
Conglomerate Dom. t	0.017 (0.033)	0.024 (0.022)	−0.0022 (0.034)	−0.0045 (0.029)	−0.011 (0.029)	−0.025 (0.025)	−0.0038 (0.039)	−0.028 (0.030)
Vertical integration			YES		YES	YES	YES	YES
Merger experience			YES			YES	YES	YES
Management				YES	YES	YES		YES
Finance				YES	YES		YES	YES
Location		YES						
Observations	120	120	120	120	120	120	120	120
R square	0.30	0.38	0.39	0.42	0.43	0.47	0.41	0.48

Notes: Table reports estimations of OLS model. Dependent variable is Malmquist-Luenberger productivity index. All models contain year dummies for 2006 through 2013 and firm fixed effects. Robust standard errors are clustered on firm level. *, ** and *** indicate significance at the 10, 5% and 1% levels.

Table A.6
Medium-run impact of Horizontal, Vertical, and Conglomerate M & As on Malmquist-Luenberger index.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Horizontal CB t−2	0.020*** (0.0048)	0.026*** (0.0053)	0.021*** (0.0045)	0.014** (0.0058)	0.016** (0.0061)	0.015** (0.0056)	0.024*** (0.0036)	0.017*** (0.0050)
Horizontal Dom. t−2	0.0030 (0.012)	0.0061 (0.016)	−0.0016 (0.012)	0.00097 (0.012)	−0.0065 (0.016)	−0.0064 (0.013)	0.00011 (0.012)	−0.0047 (0.013)
Vertical CB t−2	−0.022** (0.010)	−0.022* (0.011)	−0.018** (0.0079)	−0.015 (0.012)	−0.011 (0.0095)	−0.011 (0.0095)	−0.013 (0.0089)	−0.0080 (0.0098)
Vertical Dom. t−2	−0.011 (0.015)	−0.015 (0.023)	−0.0058 (0.016)	−0.0088 (0.014)	−0.00032 (0.015)	0.00081 (0.016)	−0.0072 (0.015)	−0.0013 (0.015)
Conglomerate CB t−2	0.017 (0.017)	−0.0023 (0.027)	0.030* (0.017)	0.017 (0.017)	0.029 (0.021)	0.025 (0.019)	0.037** (0.017)	0.031 (0.020)
Conglomerate Dom. t−2	0.0056 (0.029)	−0.0029 (0.020)	0.014 (0.034)	0.00014 (0.026)	0.012 (0.029)	0.013 (0.031)	0.012 (0.034)	0.011 (0.032)
Vertical integration			YES		YES	YES	YES	YES
Merger experience			YES			YES	YES	YES
Management				YES	YES	YES		YES
Finance				YES	YES		YES	YES
Location		YES						
Observations	120	120	120	120	120	120	120	120
R square	0.28	0.41	0.36	0.37	0.41	0.42	0.39	0.44

Notes: Table reports estimations of OLS model. Dependent variable is Malmquist-Luenberger productivity index. All models contain year dummies for 2006 through 2013 and firm fixed effects. Robust standard errors are clustered on firm level. *, ** and *** indicate significance at the 10, 5% and 1% levels.

Table A.7

Medium-run impact of domestic and cross-border mergers on Malmquist-Luenberger index.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CB mergers t–2	0.0079 (0.0049)	0.011* (0.0052)	0.011** (0.0044)	0.0060 (0.0052)	0.0088* (0.0044)	0.0082 (0.0048)	0.014*** (0.0039)	0.011** (0.0045)
Dom. mergers t–2	0.00025 (0.0073)	–0.0017 (0.0073)	0.0023 (0.0086)	–0.0014 (0.0065)	0.00095 (0.0083)	0.0015 (0.0082)	0.0022 (0.0086)	0.0012 (0.0083)
Vertical integration			YES		YES	YES	YES	YES
Merger experience			YES			YES	YES	YES
Management				YES	YES	YES		YES
Finance				YES	YES		YES	YES
Location		YES						
Observations	120	120	120	120	120	120	120	120
R square	0.23	0.35	0.32	0.35	0.39	0.40	0.35	0.42

Notes: Table reports estimations of OLS model. Dependent variable is Malmquist-Luenberger productivity index. All models contain year dummies for 2006 through 2013 and firm fixed effects. Robust standard errors are clustered on firm level. *, ** and *** indicate significance at the 10, 5% and 1% levels.

References

- Agrawal, A., Jaffe, J.F., Mandelker, G.N., 1992. The post-merger performance of acquiring firms: a re-examination of an anomaly. *J. Financ.* 47, 1605–1621.
- Akhigbe, A., Martin, A.D., 2000. Information-signaling and competitive effects of foreign acquisitions in the U.S. *J. Bank. Financ.* 24, 1307–1321.
- Aktas, N., Croci, E., Petmezas, D., 2015. Is working capital management value-enhancing? Evidence from firm performance and investments. *J. Corp. Financ.* 30, 98–113.
- Alfaro, L., Charlton, A., 2009. Intra-industry foreign direct investment. *Am. Econ. Rev.* 99, 2096–2119.
- Alfaro, L., Conconi, P., Fadinger, H., Newman, A.F., 2016. Do prices determine vertical integration? *Rev. Econ. Stud.* 83, 855–888.
- Alvarez, I., Barbero, J., Prieto, J.L.Z., 2016. A Data Envelopment Analysis toolbox for Matlab. Working Papers in Economic Theory No. 2016/03. Universidad Autónoma de Madrid (Spain), Department of Economic Analysis (Economic Theory and Economic History).
- Aparicio, J., Pastor, J.T., Zofio, J.L., 2013. On the inconsistency of the Malmquist Luenberger index. *Eur. J. Oper. Res.* 229, 738–742.
- Aparicio, J., Pastor, J.T., Prieto, J.L.Z., 2015. How to properly decompose economic efficiency using technical and allocative criteria with non-homothetic DEA technologies. *Eur. J. Oper. Res.* 240, 882–891.
- Arabi, B., Munisamy, S., Emrouznejad, A., Shadman, F., 2014. Power industry restructuring and eco-efficiency changes: a new slacks-based model in MalmquistLuenberger Index measurement. *Energy Policy* 68, 132–145.
- Asquith, P., 1983. Merger bids, uncertainty, and stockholder returns. *J. Financ. Econ.* 11, 51–83.
- Atalay, E., Hortaçsu, A., Syverson, C., 2014. Vertical Integration and Input Flows. *Am. Econ. Rev.* 104, 1120–1148.
- Atkinson, S.E., Tsionas, M.G., 2016. Directional distance functions: optimal endogenous directions. *J. Econ.* 190, 301–314.
- Aw, M.S.B., Chatterjee, R.A., 2004. The performance of UK firms acquiring large cross-border and domestic takeover targets. *Appl. Financ. Econ.* 14, 337–349.
- Aybar, B., Fici, A., 2009. Cross-border acquisitions and firm value: an analysis of emerging-market multinationals. *J. Int. Bus. Stud.* 40, 1317–1338.
- Bagdadiglu, N., Price, C.W., Weyman-Jones, T., 2007. Measuring potential gains from mergers among electricity distribution companies in Turkey using a non-parametric model. *Energy J.* 28, 83–110.
- Bai-Chen, X., Ying, F., Qian-Qian, Q., 2012. Does generation form influence environmental efficiency performance? An analysis of China's power system. *Appl. Energy* 96, 261–271.
- Balsvik, R., Haller, S.A., 2010. Picking “Lemons or Picking” Cherries? Domestic and foreign acquisitions in norwegian manufacturing. *Scand. J. Econ.* 112, 361–387.
- Baños-Caballero, S., García-Teruel, P.J., Martínez-Solano, P., 2012. How does working capital management affect the profitability of Spanish SMEs? *Small Bus. Econ.* 39, 517–529.
- Becher, D.A., Mulherin, J.H., Walking, R.A., 2012. Sources of gains in corporate mergers: refined tests from a neglected industry. *J. Financ. Quant. Anal.* 47, 57–89.
- Becker-Blease, J.R., Goldberg, L.G., Kaen, F.R., 2008. Mergers and acquisitions as a response to the deregulation of the electric power industry: value creation or value destruction? *J. Regul. Econ.* 33, 21–53.
- Bednarczyk, T.P., Schiereck, D., Walter, H.N., Walter, H.N., 2010. Cross-border acquisitions and shareholder wealth: evidence from the energy and industry in Central and Eastern Europe. *J. East Eur. Manag. Stud.*, 106–127.
- Beltrán-Estevé, M., Picazo-Tadeo, A.J., 2015. Assessing environmental performance trends in the transport industry: eco-innovation or catching-up? *Energy Econ.* 51, 570–580.
- Beltrán-Estevé, M., Picazo-Tadeo, A.J., 2017. Assessing environmental performance in the European Union: eco-innovation versus catching-up. *Energy Policy* 104, 240–252.
- Berg, S.A., Forsund, F.R., Jansen, E.S., Berg, S.A., Forsund, F.R., 1992. Malmquist Indices of Productivity Growth during the Deregulation of Norwegian Banking, 1980–89. *Scand. J. Econ.* 94, S211–S228.
- Berry, S.K., 2000. Excess returns in electric utility mergers during transition to competition. *J. Regul. Econ.* 18, 175–188.
- Bertrand, O., Betschinger, M.A., 2012. Performance of domestic and cross-border acquisitions: empirical evidence from Russian acquirers. *J. Comp. Econ.* 40, 413–437.
- Bhagat, S., Malhotra, S., Zhu, P., 2011. Emerging country cross-border acquisitions: characteristics, acquirer returns and cross-sectional determinants. *Emerg. Mark. Rev.* 12, 250–271.
- Blonigen, B.A., Fontagné, L., Sly, N., Toubal, F., 2014. Cherries for sale: the incidence and timing of cross-border M & A. *J. Int. Econ.* 94, 341–357.
- Boateng, A., Qian, W., Tianle, Y., 2008. Cross-border M & As by Chinese firms: an analysis of strategic motives and performance. *Thunderbird Int. Bus. Rev.* 50, 259–270.
- Breinlich, H., 2008. Trade liberalization and industrial restructuring through mergers and acquisitions. *J. Int. Econ.* 76, 254–266.
- Carr, D.L., Markusen, J.R., Maskus, K.E., 2001. Estimating the knowledge-capital model of the multinational enterprise. *Am. Econ. Rev.* 91, 693–708.
- Celen, A., 2013a. Efficiency and productivity (TFP) of the Turkish electricity distribution companies: an application of two-stage (DEA & Tobit) analysis. *Energy Policy* 63, 300–310.
- Celen, A., 2013b. The effect of merger and consolidation activities on the efficiency of electricity distribution regions in Turkey. *Energy Policy* 59, 674–682.
- Chari, A., Ouimet, P.P., Tesar, L.L., 2010. The value of control in emerging markets. *Rev. Financ. Stud.* 23, 1741–1770.
- Charnes, A., Cooper, W.W., Rhodes, E., 1978. Measuring the efficiency of decision making units. *Eur. J. Oper. Res.* 2, 429–444.
- Chen, Y.Y., Young, M.N., 2010. Cross-border mergers and acquisitions by Chinese listed companies: a principal-principal perspective. *Asia Pac. J. Manag.* 27, 523–539.
- Chung, Y.H., Färe, R., Grosskopf, S., 1997. Productivity and undesirable outputs: a directional distance function approach. *J. Environ. Manag.* 51, 229–240.
- Coricelli, F., Driffield, N., Pal, S., Roland, I., 2012. When does leverage hurt productivity growth? A firm-level analysis. *J. Int. Money Financ.* 31, 1674–1694.
- Datta, S., Kodwani, D., Viney, H., 2013. Shareholder wealth creation following M & A: evidence from European utility sectors. *Appl. Financ. Econ.* 23, 891–900.
- Denis, D.J., Denis, D.K., Yost, K., 2002. Global diversification, industrial diversification, and firm value. *J. Financ.* 57, 1951–1979.
- Duan, N., Guo, J.P., Xie, B.C., 2016. Is there a difference between the energy and CO₂ emission performance for China's thermal power industry? A bootstrapped directional distance function approach. *Appl. Energy* 162, 1552–1563.
- Eckbo, B.E., Thorburn, K.S., 2000. Gains to bidder firms revisited: domestic and foreign acquisitions in Canada. *J. Financ. Quant. Anal.* 35, 1–25.
- European Commission, 2004. Guidelines on the assessment of horizontal mergers under the Council Regulation on the control of concentrations between undertakings. *Off. J. Eur. Union C* 31, pp. 5–18.
- European Commission, 2008. Guidelines on the assessment of non-horizontal mergers under the Council Regulation on the control of concentrations between undertakings. *Off. J. Eur. Union C* 265, pp. 6–25.
- Färe, R., Grosskopf, S., 2004. Modeling undesirable factors in efficiency evaluation: comment. *Eur. J. Oper. Res.* 157, 242–245.
- Färe, R., Grosskopf, S., Lovell, C.A.K., Pasurka, C., 1989. Multilateral productivity comparisons when some outputs are undesirable: a nonparametric approach. *Rev. Econ. Stat.* 71, 90–98.
- Färe, R., Grosskopf, S., Pasurka, C.A., 2014. Potential gains from trading bad outputs: the case of U.S. electric power plants. *Resour. Energy Econ.* 36, 99–112.
- Fernández-Val, I., 2009. Fixed effects estimation of structural parameters and marginal effects in panel probit models. *J. Econ.* 150, 71–85.
- Fiorio, C.V., Florio, M., Doronzo, R., 2007. The electricity industry reform paradigm in the European Union: testing the impact on consumers. *Work. Pap. Berkley Electron. Press*, pp. 231–244.
- Francis, B.B., Hasan, I., Sun, X., 2008. Financial market integration and the value of

- global diversification: evidence for US acquirers in cross-border mergers and acquisitions. *J. Bank. Financ.* 32, 1522–1540.
- García-Teruel, P.J., Martínez-Solano, P., 2007. Effects of working capital management on SME profitability. *Int. J. Manag. Financ.* 3, 164–177.
- Greene, W., 2004. The behaviour of the maximum likelihood estimator of limited dependent variable models in the presence of fixed effects. *Econ. J.* 7, 98–119.
- Gubbi, S.R., Aulakh, P.S., Ray, S., Sarkar, M.B., Chittoor, R., 2010. Do international acquisitions by emerging-economy firms create shareholder value? The case of Indian firms. *J. Int. Bus. Stud.* 41, 397–418.
- Guerreiro, C., de Leeuw, F., Foltescu, V., Horálek, J., 2014. Air quality in Europe - 2014 report. EEA Report No 5/2014. European Environment Agency.
- Hahn, J., Newey, W., 2004. Jackknife and analytical bias reduction for nonlinear panel models. *Econometrica* 72, 1295–1319.
- Hattori, T., Jamasb, T., Pollitt, M., 2005. Electricity distribution in the UK and Japan: a comparative efficiency analysis 1985–1998. *Energy J.*, 23–47.
- Helman, U., 2006. Market power monitoring and mitigation in the US wholesale power markets. *Energy* 31, 877–904.
- Herger, N., McCorriston, S., 2016. Horizontal, vertical, and conglomerate cross-border acquisitions. *IMF Econ. Rev.* 64, 319–353.
- Himmelberg, C.P., Hubbard, G.R., Palia, D., 1999. Understanding the determinants of managerial ownership and the link between ownership and performance. *J. Financ. Econ.* 53, 353–384.
- Jamasb, T., Pollitt, M., 2003. International benchmarking and regulation: an application to European electricity distribution utilities. *Energy Policy* 31, 1609–1622.
- Jensen, M.C., Ruback, R.S., 1983. The market for corporate control: the scientific evidence. *J. Financ. Econ.* 11, 5–50.
- Jose, M.L., Lancaster, C., Stevens, J.L., 1996. Corporate returns and cash conversion cycles. *J. Econ. Financ.* 20, 33–46.
- Kamien, M.I., Zang, I., 1993. Monopolization by sequential acquisition. *J. Law Econ. Organ.* 9, 205–229.
- Kao, C., Liu, S.T., 2014. Multi-period efficiency measurement in data envelopment analysis: the case of Taiwanese commercial banks. *Omega* 47, 90–98.
- King, D.R., Dalton, D.R., Daily, C.M., Covin, J.G., 2004. Meta-analyses of post-acquisition performance: indications of unidentified moderators. *Strateg. Manag. J.* 25, 187–200.
- Kiyamaz, H., Mukherjee, T.K., 2000. The impact of country diversification on wealth effects in cross-border mergers. *Financ. Rev.* 35, 37–58.
- Kraus, A., Litzenberger, R.H., 1973. A state-preference model of optimal financial leverage. *J. Financ.* 28, 911–922.
- Kwoka, J., Pollitt, M., 2010. Do mergers improve efficiency? Evidence from restructuring the US electric power sector. *Int. J. Ind. Organ.* 28, 645–656.
- Kwoka, J.E., 2002. Vertical economies in electric power: evidence on integration and its alternatives: evidence on integration and its alternatives. *Int. J. Ind. Organ.* 20, 653–671.
- Leggio, K.B., Lien, D., 2000. Mergers in the electric utility industry in a deregulatory environment. *J. Regul. Econ.* 17, 69–85.
- Lewellen, W.G., 1971. A pure financial rationale for the conglomerate merger. *J. Financ.* 26, 521–537.
- Lichtenberg, F.R., Siegel, D., Jorgenson, D., Mansfield, E., 1987. Productivity and changes in ownership of manufacturing plants. *Pap. Econ. Act.* 1987, 643–683.
- Loderer, C., Martin, K., 1992. Postacquisition performance of acquiring firms. *Financ. Manag.* 21, 69–79.
- Markides, C.C., Iltner, C.D., 1994. Shareholder benefits from corporate international diversification: evidence from U.S. international acquisitions. *J. Int. Bus. Stud.* 25, 343–366.
- Martynova, M., Renneboog, L., 2008. Spillover of corporate governance standards in cross-border mergers and acquisitions. *J. Corp. Financ.* 14, 200–223.
- McDonald, J., 2009. Using least squares and tobit in second stage DEA efficiency analyses. *Eur. J. Oper. Res.* 197, 792–798.
- McGuckin, R.H., Nguyen, S.V., 1995. On productivity and plant ownership change: new evidence from the Longitudinal Research Database. *RAND J. Econ.* 26, 257–276.
- Moeller, S.B., Schlingemann, F.P., Stulz, R.M., 2004. Firm size and the gains from acquisitions. *J. Financ. Econ.* 73, 201–228.
- Moeller, S.B., Schlingemann, F.P., Stulz, R.M., 2005. Wealth destruction on a massive scale? A study of acquiring-firm returns in the recent merger wave. *J. Financ.* 60, 757–782.
- Murteira, J.M.R., Ramalho, J.J.S., 2016. Regression analysis of multivariate fractional data. *Econom. Rev.* 35, 515–552.
- Neary, P.J., 2007. Cross-border mergers as instruments of comparative advantage. *Rev. Econ. Stud.* 74, 1229–1257.
- Nocke, V., Yeaple, S., 2007. Cross-border mergers and acquisitions vs. greenfield foreign direct investment: the role of firm heterogeneity. *J. Int. Econ.* 72, 336–365.
- Papke, L.E., Wooldridge, J.M., 1996. Econometric methods for fractional response variables with an application to 401(k) plan participation rates. *J. Appl. Econ.* 11, 619–632.
- Papke, L.E., Wooldridge, J.M., 2008. Panel data methods for fractional response variables with an application to test pass rates. *J. Econ.* 145, 121–133.
- Pérez-Reyes, R., Tovar, B., 2009. Measuring efficiency and productivity change (PTF) in the Peruvian electricity distribution companies after reforms. *Energy Policy* 37, 2249–2261.
- Perrin, L.M., 2013. Mapping power and utilities regulation in Europe. Technical Report. Assurance Power and Utilities Sector Resident, EY.
- Pollitt, M.G., 2009. Electricity Liberalisation in the European Union: A Progress Report. Cambridge Working Paper in Economics No. 0953. University of Cambridge.
- Pombo, C., Tabora, R., 2006. Performance and efficiency in Colombia's power distribution system: effects of the 1994 reform. *Energy Econ.* 28, 339–369.
- PricewaterhouseCoopers, 2013. Climate Change and Electricity: European Carbon Factor, benchmarking of CO2 emissions by Europe's largest electricity utilities. *Les Cahiers du Développement Durable* 12th Edition. PwC France.
- Rahman, M., Lambkin, M., Hussain, D., 2016. Value creation and appropriation following M & A: a data envelopment analysis. *J. Bus. Res.* 69, 5628–5635.
- Ramalho, E.A., Ramalho, J.J.S., Murteira, J.M.R., 2011. Alternative estimating and testing empirical strategies for fractional regression models. *J. Econ. Surv.* 25, 19–68.
- Ramondo, N., Rappoport, V., Ruhl, K.J., 2016. Intrafirm trade and vertical fragmentation in U.S. multinational corporations. *J. Int. Econ.* 98, 51–59.
- Saastamoinen, A., Bjørndal, E., Bjørndal, M., 2017. Specification of merger gains in the Norwegian electricity distribution industry. *Energy Policy* 102, 96–107.
- Sahoo, B.K., Luptack, M., Mahlberg, B., 2011. Alternative measures of environmental technology structure in DEA: an application. *Eur. J. Oper. Res.* 215, 750–762.
- Schmid, A.S., Sánchez, C.M., Goldberg, S.R., 2012. M & A today: great challenges, but great opportunities. *J. Corp. Account. Financ.* 23, 3–8.
- Serrallés, R.J., 2006. Electric energy restructuring in the European Union: integration, subsidiarity and the challenge of harmonization. *Energy Policy* 34, 2542–2551.
- Servaes, H., Zenner, M., 1994. Taxes and the returns to foreign acquisitions in the United States. *Financ. Manag.* 23, 42–56.
- Shimizu, K., Hitt, M.A., Vaidyanath, D., Pisano, V., 2004. Theoretical foundations of cross-border mergers and acquisitions: A review of current research and recommendations for the future. *J. Int. Manag.* 10, 307–353.
- Simar, L., Wilson, P.W., 2007. Estimation and inference in two-stage, semi-parametric models of production processes. *J. Econ.* 136, 31–64.
- Sözen, A., Alp, H., Özdemir, A., 2010. Assessment of operational and environmental performance of the thermal power plants in Turkey by using data envelopment analysis. *Energy Policy* 38, 6194–6203.
- Steiner, F., 2001. Regulation, industry structure, and performance in the electricity supply industry. *OECD Econ. Stud.* 32, 143–182.
- Sueyoshi, T., Goto, M., Ueno, T., 2010. Performance analysis of US coal-fired power plants by measuring three DEA efficiencies. *Energy Policy* 38, 1675–1688.
- Sueyoshi, T., Yuan, Y., Goto, M., 2017. A literature study for DEA applied to energy and environment. *Energy Econ.* 62, 104–124.
- Sueyoshi, T., Yuan, Y., 2016. DEA Environmental Assessment (II): A Literature Study. In: *Handb. Oper. Anal. Using Data Envel. Anal.*. Springer, pp. 445–481.
- Vaninsky, A., 2006. Efficiency of electric power generation in the United States: Analysis and forecast based on data envelopment analysis. *Energy Econ.* 28, 326–338.
- Wang, Q., Zhao, Z., Zhou, P., Zhou, D., 2013. Energy efficiency and production technology heterogeneity in China: a meta-frontier DEA approach. *Econ. Model.* 35, 283–289.
- Wang, Y.J., 2002. Liquidity management, operating performance, and corporate value: evidence from Japan and Taiwan. *J. Multinat. Financ. Manag.* 12, 159–169.
- Wooldridge, J.M., 2009. Correlated random effects models with unbalanced panels. Manuscript (Version July 2009). Department of Economics, Michigan State University.
- Yang, H., Pollitt, M., 2009. Incorporating both undesirable outputs and uncontrollable variables into DEA: The performance of Chinese coal-fired power plants. *Eur. J. Oper. Res.* 197, 1095–1105.
- Zhang, N., Choi, Y., 2014. A note on the evolution of directional distance function and its development in energy and environmental studies 1997–2013. *Renew. Sustain. Energy Rev.* 33, 50–59.
- Zhang, N., Zhou, P., Choi, Y., 2013. Energy efficiency, CO2 emission performance and technology gaps in fossil fuel electricity generation in Korea: a meta-frontier non-radial directional distance function analysis. *Energy Policy* 56, 653–662.
- Zhou, P., Ang, B.W., 2008. Linear programming models for measuring economy-wide energy efficiency performance. *Energy Policy* 36, 2911–2916.
- Zhou, P., Ang, B.W., Poh, K.L., 2008. A survey of data envelopment analysis in energy and environmental studies. *Eur. J. Oper. Res.* 189, 1–18.
- Zhou, P., Ang, B.W., Han, J.Y., 2010. Total factor carbon emission performance: a Malmquist index analysis. *Energy Econ.* 32, 194–201.
- Zhou, P., Ang, B.W., Wang, H., 2012. Energy and CO2 emission performance in electricity generation: a non-radial directional distance function approach. *Eur. J. Oper. Res.* 221, 625–635.